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Department of Statistics

Virtual University of Pakistan

STA304



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Lecture 1

Index Number

Selection of Base Period

Fixed Base Method

Chain Base Method

Selection of Average

Index Numbers:

An index number is a statistical measure of average change in a variable or group of variables with respect to time or space.

The classical definition of an index number is provided by the English economist F.Y.Edgeworth (1854 – 1926) who states that "An index number is a quantity which shows by its variations, the changes over time or space of a magnitude which is not subject either of accurate measurement in itself or of direct valuation in practice".

Example:

Suppose that a cup of coffee in a particular café cost Rs. 35 in 2002. In 2012, an identical cup of coffee cost Rs. 50. How has the price changed from 2002 to 2012?

Construction of index numbers

The usual method of compilation of an index number of wholesale prices involves the following steps:

- Selection of commodities to be included, their number and price quotations.
- Selection of the base period and calculation of price relatives.
- Selection of average to be used.
- Selection of appropriate weights.

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Selection of Commodities

The following steps are included for selection of commodities:

1. A reasonable number of commodities on the basis of their evaluated importance should be used.
2. While deciding on their number, the commodities to be selected must be taken as,

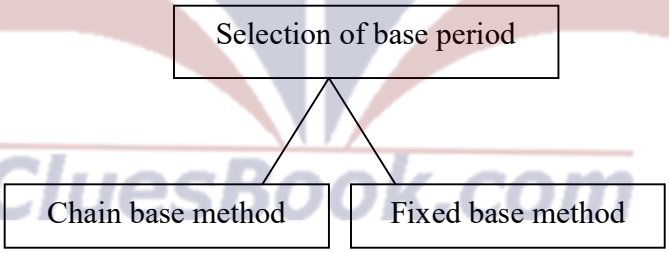
i. Representative of the tastes, habits and requirements of the people concerned

ii. Unlikely to vary in quality or grade

iii. Comparable

Selection of Base Period

There are two methods for selection of base period.



Notations

- p_t :prices (of a vector of commodities) at time t
- q_t :real quantities purchase (of a vector of commodities) at time t
- Suppose 2 periods
  - Base period: t=0
  - Current period: t=1

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Fixed Base Method

A fixed base method is one in which a particular year is generally chosen as the base period that remains unchanged during the life term of the index.

Example:

Suppose that a cup of coffee in a particular café cost Rs. 35 in 2002. In 2012, an identical cup of coffee cost Rs. 50. How has the price changed from 2002 to 2012?

The particular time period of 1995 which we've chosen to compare against, is called the **base period**.

Price relative

A price relative expresses the price of a commodity in a given year as a fraction of the price in the base year. It is multiplied by 100 to make it a percentage.

$$\text{Price relative} = \frac{P_n}{P_0} \times 100$$

Where P_o denotes the price of the base year and P_n denotes the price of the current period.

Example

Suppose that a cup of coffee in a particular café cost Rs. 35 in 2002. In 2012, an identical cup of coffee cost Rs. 50. How has the price changed from 2002 to 2012?

The particular time period of 2002 which we've chosen to compare against, is called the **base period**.

Here Po = 35, Pn = 50, then the

$$\text{Price relative} = \frac{P_n}{P_0} \times 100$$

$$= \frac{50}{35} \times 100$$

$$= 142\%.$$

Example

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Find the price relatives for each year from the following average retail prices of wheat, using

(i) 1999 as a base

(ii) 2005 as a base

Year	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008
Retail Price	16	18	18	20	23	25	40	70	90	150

Solution

Year	**Retail Prices (Rs.)**	**Price Relatives**	
		1999 as base	**2005 as base**
1999	16	100	40
2000	18	112.5	45
2001	18	112.5	45
2002	20	125	50
2003	23	143.8	57.5
2004	25	156.3	62.5
2005	40	250	100
2006	70	437.5	175
2007	90	562.5	225
2008	150	937.5	375

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Chain Base Method

A chain base method is one in which the base period is not fixed, but moves with the given year. For example, for 2008, 2007 will be the base; for 2007,2006 will be the base and so on.

Example

Suppose that a cup of coffee in a particular café cost Rs. 35 in 2002. In 2012, an identical cup of coffee cost Rs. 50. In 2014, an identical cup of coffee cost Rs. 60.

Link Relative

The relatives are computed with the immediately preceding year as the base and such relatives are called link relatives.

$$\frac{P_n}{P_{n-1}} \times 100$$

Where P_n denotes the price of the commodity in the given year and P_{n-1} denotes the price of the commodity in the preceding year.

Example

From the data given below construct an index number by chain base method.

Price of a commodity from 2006 to 2008.

Year	**Price**
2006	50
2007	60
2008	65

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Solution

Year	Price	Link Relative
2006	50	100
2007	60	$\frac{60}{50} * 100 = 120$
2008	65	$\frac{65}{60} * 100 = 108$

Example

The price relatives of three commodities are given below:

Year	Price Relatives of Commodities		
	A	B	C
1934	100	100	100
1935	105	97	121
1936	110	94	125
1937	115	100	130
1938	116	99	128
1939	120	105	130

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Compute the link relative, i.e. the price relatives in each year with reference to the previous year as 100.

Solution

Year	Link Relatives of Commodities		
	A	B	C
1934	100	100	100
1935	105.0	97.0	121.0
1936	104.8	96.9	103.3
1937	104.5	106.4	104.0
1938	100.9	99.0	98.5
1939	103.5	106.1	101.6

Selection of Average

The choice of an appropriate average to get a single index number for each year. For this purpose, any of the following averages may be used:

1. The arithmetic mean
2. The median
3. The geometric mean

Selection of Appropriate Weights

The construction of wholesale price index numbers is to decide how to select weights which would indicate the relative importance various commodities in the group. It is evident that all the commodities selected are not equally important.

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For example, eggs and coffee cannot be given the same importance as wheat and rice. Wheat is much more important than coffee, it is therefore, desirable that wheat must be given more importance of each commodity, a sample survey should be conducted.

There are two type of indices.

- Un weighted indices- in which no specific weights are attached
- Weighted indices- in which appropriate weights are assigned to various items.

Example

The following table gives the production of cotton cloth (in million yards) for Pakistan from 1954 to 1963.

Year	1954	1955	1956	1957	1958	1959	1960	1961	1962	1963
Production	282	389	438	470	511	555	564	630	662	681

Find out the index numbers by taking:

(i) 1954 as base year

(ii) Average of 1958, 59, 60 as base period.

Solution

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Year	Production	Price Relatives	
		1954 as base	Average of 1958, 1959 and 1960 as base
1954	282	100	51.90
1955	389	137.94	71.60
1956	438	155.32	80.61
1957	470	166.67	86.50
1958	511	181.21	94.05

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1959	555	196.81	102.15
1960	564	200.00	103.80
1961	630	223.40	115.95
1962	662	234.75	121.84
1963	681	241.49	125.34

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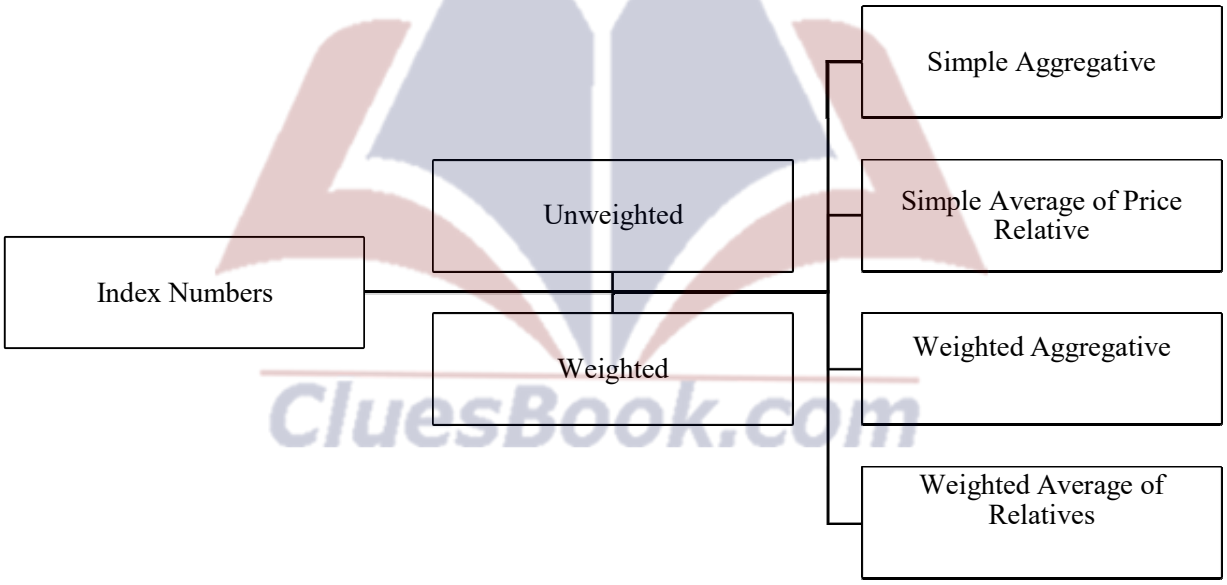
Lecture 2

Method of Construction of Index Numbers

Simple Aggregative Method

Simple Average of Relatives

METHOD OF CONSTRUCTION OF INDEX NUMBERS



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Unweighted Index Numbers

Simple Aggregative Method

It consists in expressing the aggregate price of all commodities in the current year as a percentage of the aggregate price in the base year.

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$$P_{0n} = \frac{\sum p_n}{\sum p_0} \times 100.$$

Where P_{0n} denotes the price index for the given year *n* relative to the base year 0,

- denotes the sum of prices for given year, and
- denotes the sum of prices for base year.

Example

The average prices of certain commodities are given below:

Commodity	Unit	Prices (Rs.) in	
		1957	1958
Wheat	Ton	351.00	335.00
Rice	Maund	35.00	32.00
Salt	Maund	10.00	11.00
Sugar	Seer	1.25	1.40
Cloth	Yard	2.25	2.60
Milk	Seer	0.75	0.85
Oil	Gallon	1.25	1.35

Compute a simple aggregative price index number for the year 1958 based on 1957 prices.

Solution

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Unweighted Index Numbers

Simple Average of Relatives

A simple average of price relatives is an index obtained by taking the average of the price relatives of the given commodities for each year and expressing the result as a percentage. If we take the arithmetic mean, then we have

$$P_{0n} = \frac{1}{k}\sum\left(\frac{P_n}{P_0}\right)\times 100.$$

Where k denotes the number of commodities whose price relatives are thus combined.

When geometric mean is used:

$$P_{0n} = \left[\Pi\left(\frac{P_n}{P_0}\right)\right]^{1/k}\times 100,$$

Example

The following table gives the average wholesale prices in rupees per unit of gold, wheat and cotton during the years 1912 – 1917.

Commodities	Average prices in Rs. Per unit					
	1912	1913	1914	1915	1916	1917
Gold	25.3	30.8	33.4	35.5	35.3	36.0
Wheat	17.3	14.5	4.9	5.7	17.1	11.6
Cotton	7.8	5.4	6.7	5.6	7.2	10.2

Using 1912 as the base period, compute the simple average of relatives price indices for the years 1913 to 1917.

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Solution

Years	Price relatives			Sum of price relatives	Index No. as average of relatives
	Gold	Wheat	Cotton		
1912	100	100	100	300	100
1913	122	84	69	275	91.6
1914	132	28	86	246	82.1
1915	140	33	72	245	81.7
1916	140	100	92	332	110.7
1917	142	67	131	340	113.4

Years	Price relatives			Sum of price relatives	Index No. as average of relatives
	Gold	Wheat	Cotton		
1912					
1913					
1914					
1915					
1916					
1917					

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Simple Average of Price Relatives

In the previous lecture, we read the method of simple average of price relatives by using the three different methods:

- Mean
- Median
- Geometric Mean

Example

From the data given below, compute the index number of prices, taking 1962 as base. Use

(i) simple average of price relatives.

(ii) the median of price relatives.

(iii) the geometric mean of price relatives.

	Commodity (Price in Rs.)			
Year	Firewood	Soft cake	Kerosene oil	Matches
1962	3.25	2.50	0.20	0.06
1963	3.44	2.80	0.22	0.06
1964	3.50	2.00	0.25	0.06
1965	3.75	2.50	0.25	0.06

Solution

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	Price Relatives				**Total**	**Index Numbers by**	
Year	Firewood	Soft cake	Kerosene oil	Matches		(i) Mean	(ii) Median
1962	100	100	100	100	400	100	100
1963	106	112	110	100	428	107	108
1964	108	80	125	100	413	103	104
1965	115	100	125	100	440	110	108

	Price Relatives				$\sum \log P$	$\frac{1}{k}\sum \log P$	G.M
Year	Firewood	Soft cake	Kerosene oil	Matches			
1962	100	100	100	100	8	2	100
Log x	2	2	2	2			
1963	106	112	110	100	8.116	2.029	106.90
Log x	2.025	2.049	2.041	2			
1964	108	80	125	100	8.033	2.008	101.86
Log x	2.033	1.903	2.097	2			
1965	115	100	125	100	8.158	2.039	109.39
Log x	2.061	2	2.097	2			

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Chain Indices

Chain indices are the index numbers which are calculated by using link relatives.

Example

Construct chain indices for the following years, taking 1940 as the base and using the simple average of relatives.

Year	Price in Rs. Per mound		
	Wheat	Rice	Maize
1940	2.80	10.50	2.70
1941	3.40	10.80	3.20
1942	3.60	10.60	3.50
1943	4.00	11.00	3.80
1944	4.20	11.50	4.00

Solution

First, we calculate the link relatives by the formula:

$$\text{Link relative} = \frac{Price\ of\ a\ commodity\ in\ the\ given\ year}{Price\ of\ the\ commodity\ in\ the\ preceding\ year} \times 100$$

$$\text{Or } P_{n-1,n} = \frac{P_n}{P_{n-1}} \times 100$$

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Year	Iink Relatives			Sum of relatives	Simple average of relatives	Chain Indices
	Wheat	Rice	Maize			
1940	100	100	100	300	100	100
1941	121	103	119	343	114	114
1942	106	98	109	313	104	118.6
1943	111	104	109	324	108	128.1
1944	105	105	105	315	105	134.5

Example

Find the chain indices from the following price relatives of four commodities, using the

- Geometric mean of the relatives for each year.

	Commodity			
Year	A	B	C	D
1951	81	77	119	55
1952	62	104	87	82
1953	104	87	111	100
1954	93	75	154	96
1955	60	43	165	88

Solution

The link relatives for commodity A (for instance) are computed below:

Link relative for 1952 $= \frac{62}{81} \times 100 = 76$

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Link relative for 1953 = $\frac{104}{62} \times 100 = 168$

Link relative for 1954 = $\frac{93}{104} \times 100 = 89$

Link relative for 1952 = $\frac{62}{81} \times 100 = 76$

Calculation of chain indices, using the geometric mean of the relatives.

	Link Relatives						G.M.	Chain Index Numbers
Year	A	B	C	D				
1951	81	77	119	55	7.6100	1.9027	79.9	79.9
1952	76	70	108	149	7.9325	1.9831	96.2	76.9
1953	168	161	87	122	8.4580	2.1145	130.2	100.1
1954	89	86	139	96	8.0092	2.0023	100.6	100.7
1955	65	57	107	93	7.5667	1.8917	77.9	78.4

Hence the chain indices by geometric mean are 79.91 76.9, 100.1, 100.7 and 78.4.

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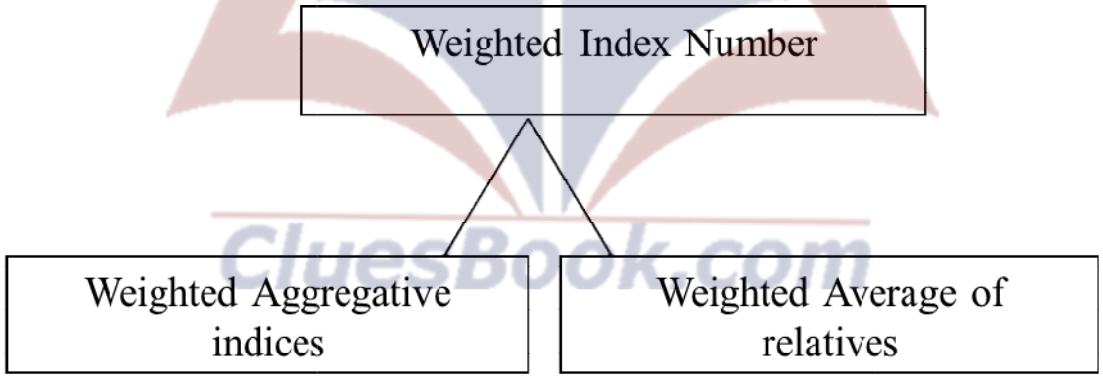
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Lecture 3

- Weighted Index Numbers
- Weighted Aggregative Price Index Numbers
- Weighted Average of Relatives Price Index Numbers

Weighted Index Numbers

An index number that measures the change in the prices of a group of commodities when the relative importance of the commodities (i.e. Weight) has been taken into account, is called a weighted price index number. Weighted indices are generally divided into Weighted Aggregative indices and Weighted Average of relatives indices.

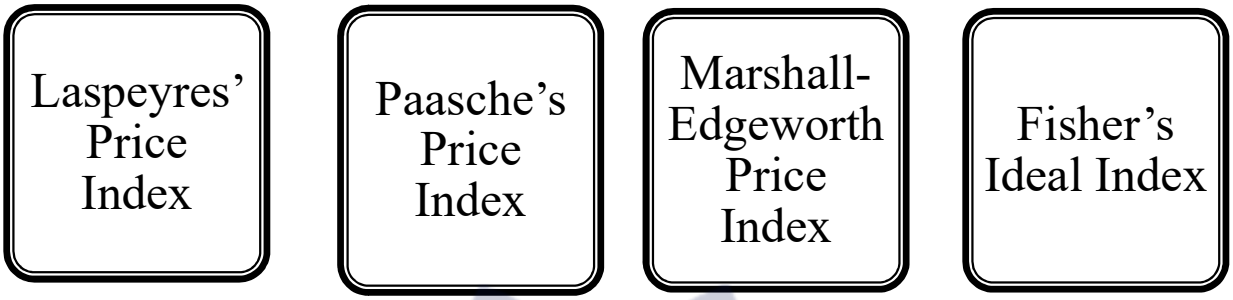


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Weighted Aggregative Price Index Numbers

These index numbers are the simple aggregative type with the fundamental difference that weights are assigned to the various items included in the index. There are various kinds of weighted aggregative index numbers are,



Laspeyres' Price Index

This method was derived by Laspeyres' in 1871. In this method the weights are determined by quantities in the base.

$$P_{0n} = \frac{\sum p_n q_0}{\sum p_0 q_0} \times 100$$

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Paasche's Price Index

This method was derived by German statistician Paasche's in 1874. The weights of current year are used as base year in constructing the Paasche's Index number.

$$P_{0n} = \frac{\sum p_n q_n}{\sum p_0 q_n} \times 100$$

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Example

Construct the following weighted aggregative price index numbers for 1960 by using 1956 as a base period.

- Laspeyres' index
- Paasche's index

Commodity	Prices (Rs. Per md.)		Quantities (tons)	
	1956 (base)	1960	1956	1960
A	64	75	270	276
B	40	45	124	118
C	18	21	130	121
D	58	68	185	267

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Solution

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Com.	Price		Quantity		p_0q_0	p_1q_0	p_0q_1	p_1q_1
	p_0	p_1	q_0	q_1				
A	64	75	270	276	17280	20250	17664	20700
B	40	45	124	118	4960	5580	720	5310
C	18	21	130	121	2340	2730	2178	2541
D	58	68	185	267	10730	12580	15486	18156
Total					35310	41140	40048	46707

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(i) **Laspeyres' index numbers**

Index for 1960 $= \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 = \frac{41140}{35310} \times 100 = 116.5$

(ii) **Paasche's index numbers**

Index for 1960 $= \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100 = \frac{46707}{40048} \times 100 = 116.6$

Marshall-Edgeworth Price Index

In this index the numerator consists of an aggregate of the current years price multiplied by the weights of both of the base year as well as the current year.

$$P_{0n} = \frac{\sum p_n (q_0 + q_n)}{\sum p_0 (q_0 + q_n)}$$

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Fisher's Ideal Index Number

Fisher's ideal index number is the geometric mean of the Laspeyres' and Paasche's index numbers.

$$P_{0n(Fisher)} = \sqrt{P_{0n}(\text{Laspeyers}) P_{0n}(\text{Paasches})}$$

$$P_{0n} = \sqrt{\frac{\sum p_n q_0}{\sum p_0 q_0} \times \frac{\sum p_n q_n}{\sum p_0 q_n}} \times 100$$

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Example

The prices and quantities of three commodities during 1960 and 1964 are given below:

Commodity	Prices (Rs. Per md.)		Quantities Produced (maunds)	
	1960 (p_o)	**1964 (p_1)**	**1960 (q_o)**	**1964 (q_1)**
A	3.95	4.25	9,676	10,436
B	34.80	38.94	78	83
C	61.56	59.70	118	116

Compute the Marshall-Edgeworth index numbers and Fisher Index numbers for 1964, using 1960 as the base period.

Solution

To calculate the Marshal-Edgeworth, first we calculate the necessary products with 1960 as base year. These calculations are shown in the table below:

Computation of Weighted Aggregative Price Indices

Com.	**(p_0+q_1)**	**$p_1(q_0+q_1)$**	**$p_0(q_0+q_1)$**

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A B C	20,111 161 234	85471.75 6269.34 13,969.80	79,438.45 5,602.80 14,40.04
Total		105,710.89	99,446.29

Marshall-Edgeworth price index:

$$\frac{\sum p_1(q_0 - q_1)}{\sum p_0(q_0 + q_1)} \times 100$$

$$= \frac{105710.89}{99446.29} \times 100$$

= 106.3.

Com.	**Price**		**Quantity**		p_0q_0	p_1q_0	p_0q_1	p_1q_1
	p_0	p_1	q_0	q_1				
A	3.95	4.25	9676	10436	38220.2	41123	41222.2	44353
B	34.80	38.94	78	83	2714.4	3037.32	2888.4	3232.02
C Total	61.56	59.70	118	116	7264.08	7044.6	7140.96	6925.2
					48198.68	51204.92	51251.56	54510.22

Fisher's "Ideal" index numbers:

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$$\sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} \times 100$$

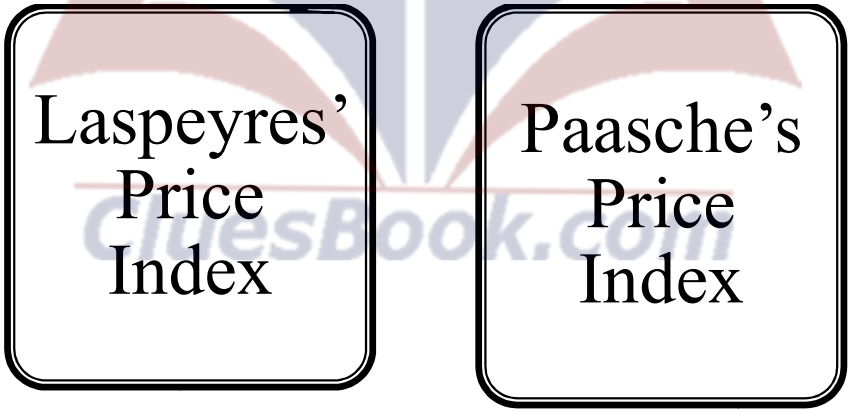
$$= \sqrt{\frac{51204.92}{48198.68} \times \frac{54510.22}{51251.56}} \times 100$$

$$= \sqrt{1.062 \times 1.064} \times 100$$

=106.30.

Weighted Average of Relatives Price Index Numbers

It is computed multiplying each price relative by its weight, summing these products and dividing by the sum of the weights. The important types of the weighted average of relatives price indices are given below:



Laspeyres' Index Numbers

$$P_{0n} = \frac{\sum (p_n / p_0) p_0 q_0}{\sum p_0 q_0} \times 100$$

Where the price relatives are weighted by the total value of commodities in the base year.

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Paasche's Index Numbers

$$P_{0n} = \frac{\sum \left(\frac{p_n}{p_0}\right) p_0 q_n}{\sum p_0 q_n} \times 100$$

Where the price relatives are weighted by the total value of commodities in the given year at base year.

Example

Using the data, compute the weighted average of relatives price index numbers for 1964 by

- Laspeyres' method
- Paasche's method

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Commodity	Prices (Rs. Per md.)		Quantities Produced (maunds)	
	1960 (p_o)	**1964 (p_1)**	**1960 (q_o)**	**1964 (q_1)**
A	3.95	4.25	9,676	10,436
B	34.80	38.94	78	83
C	61.56	59.70	118	116

Solution

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Com.	Price		Quantity		Price Relative	Weights		Weighted Price Relatives	
	p_o 1960	p_1 1964	q_o 1960	q_1 1964		p_oq_o	p_oq_1		
A	3.95	4.25	9,676	10,436	1.076	38220.2	41222.20	41124.94	44355.09
B	34.80	38.94	78	83	1.119	2714.40	2888.40	3037.41	3232.12
C	61.56	59.70	118	116	0.970	7264.08	7140.95	7046.16	6926.73
Total						48198.68	51251.56	51208.51	54513.94

Laspeyres' index for 1964 = $\frac{\sum (p_n/p_0)p_0q_0}{\sum p_0q_0} \times 100$

$$= \frac{51,208.51}{48,198.68} \times 100 = 106.24,$$

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Paasche's index for 1964 = $\frac{\sum \left(\frac{p_n}{p_0}\right) p_0q_n}{\sum p_0q_n} \times 100$

$$= \frac{54,513.94}{51,251.56} \times 100 = 106.4.$$

Example

Compute Fisher's price index numbers for 1976 with 1961 as base from the following data:

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Commodity	Quantity (units)		Value (Rs.)	
	1961	1976	1961 (p_0q_0)	1976 (p_1q_1)
A	100	150	600	1200
B	80	100	400	700
C	60	72	180	432
D	30	33	450	363
Total			1630	
			2695	

Solution

Unit prices		p_0q_1	p_1q_0
p_0	p_1		
6	8	800	900
5	7	560	500
3	6	360	216
15	11	330	495
		2050	2111

Fisher's index numbers for 1976:

$$\sqrt{\frac{\sum p_1q_0}{\sum p_0q_0} \times \frac{\sum p_1q_1}{\sum p_0q_1}} \times 100$$

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$$= \sqrt{\frac{2050}{1630} \times \frac{2695}{2111}} \times 100$$

$$= \sqrt{1.2577 \times 1.2767} \times 100$$

$$= 126.72.$$

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Lecture 4

- Tests for Index Number
- Consumer Price Index
- Construction of Consumer Price Index
- Sensitive Price Indicator

Test for Index Number

From a theoretical viewpoint, a "good" index number formula is required to satisfy the following tests proposed by Irvif Fisher (1867 – 1947).

- Time Reversal Test
- Factor Reversal Test
- Circular Test

Time Reversal Test

Time Reversal Test may be stated as follows:

"If the time subscripts 0 and n are interchanged in a price (or quantity) index number formula wherever they appear, then the resulting price (or quantity) formula should be the reciprocal of the original index formula, ignoring the factor 100." Symbolically, the test requires that

$$P_{0n} = \frac{1}{P_{n0}} \text{ or } P_{0n} \times P_{n0} = 1$$

For the purposes of illustration, let us consider some of the important formulae and see whether they or do not satisfy this test.

i. The Laspeyres' price index expressed in ratio rather than percentage form (i.e. omitting the factor 100), is given by:

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$$P_{0n} = \frac{\sum p_n q_0}{\sum p_0 q_0}$$

Interchanging the time subscripts, we get

$$P_{n0} = \frac{\sum p_0 q_n}{\sum p_n q_n}$$

But $P_{0n} \times P_{n0}$, i.e. $\frac{\sum p_n q_0}{\sum p_0 q_0} \times \frac{\sum p_0 q_n}{\sum p_n q_n} \neq 1$.

Hence it does not satisfy the time reversal test.

ii. Again, taking the Paasche's price index formula, i.e. $\frac{\sum p_n q_n}{\sum p_0 q_n}$

$$P_{n0} = \frac{\sum p_n q_n}{\sum p_0 q_n}$$

and interchanging the time subscripts, we get

But again $P_{0n} \times P_{n0}$ i.e. $\frac{\sum p_n q_n}{\sum p_0 q_n} \times \frac{\sum p_0 q_0}{\sum p_n q_0} \neq 1$, which shows that this formula, too,

does not obey the time reversal test.

iii. Interchanging the time subscripts in the Marshall-Edgeworth price index, i.e. in the formula

$$P_{0n} = \frac{\sum p_n (q_0 + q_n)}{\sum p_n (q_0 + q_n)}$$

Multiplying together, we obtain

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$$P_{0n} \times P_{n0} = \frac{\sum p_n(q_0+q_n)}{\sum p_n(q_0+q_n)} \times \frac{\sum p_0(q_n+q_0)}{\sum p_0(q_n+q_0)} = 1$$

Hence the time reversal test is satisfied.

iv. Interchanging the time subscripts in the Fisher price index, i.e. in the formula

$$P_{0n} = \sqrt{\frac{\sum p_n q_0}{\sum p_0 q_0} \times \frac{\sum p_n q_n}{\sum p_0 q_n}}$$

Multiplying together, we obtain

$$P_{0n} \times P_{n0} = \sqrt{\frac{\sum p_n q_0}{\sum p_0 q_0} \times \frac{\sum p_n q_n}{\sum p_0 q_n}} \times \sqrt{\frac{\sum p_0 q_n}{\sum p_n q_0} \times \frac{\sum p_0 q_0}{\sum p_n q_0}} = 1.$$

Hence the time reversal test is satisfied.

Factor Reversal Test

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Factor Reversal Test may be stated as follows:

"If the factors p's (prices) and q's (quantities) occurring in a price (or quantity) index formula be interchanged (or reversed) so that a quantity (or price) index formula is obtained, then the product of the two index numbers should equal the value index number, i.e. In other words, the factor reversal test requires that

(Price index) (Quantity index) = Value index.

For the purposes of illustration, let us consider some of the important formulae and see whether they or do not satisfy this test.

i. Interchanging the factors p's and q's in the formula

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$$P_{0n} = \frac{\sum p_n q_0}{\sum p_0 q_0}$$ (Laspeyres' price index)

$$Q_{0n} = \frac{\sum q_n p_0}{\sum q_0 p_0}$$ (Laspeyres' quantity index)

The product of these two is $\frac{\sum p_n q_0}{\sum p_0 q_0} \times \frac{\sum q_n p_0}{\sum q_0 p_0}$, which is not equal to $\frac{\sum p_n q_n}{\sum p_0 q_0}$, the value index.

ii. Again interchanging the factors *p's* and *q's,* the formula

$$P_{n0} = \frac{\sum p_n q_n}{\sum p_0 q_n}$$ (Paasche's price index)

transforms into $Q_{n0} = \frac{\sum q_n p_n}{\sum q_0 p_n}$ (Paasche's quantity index)

Thus $P_{0n} \times Q_{0n} = \frac{\sum p_n q_n}{\sum p_0 q_n} \times \frac{\sum q_n p_n}{\sum q_0 p_n} \neq \frac{\sum p_n q_n}{\sum p_0 q_0}$ (Value Index)

Hence Paasche's price index does not satisfy the factor reversal test.

iii. The Marshall- Edgeworth's price index is

$$P_{0n} = \frac{\sum p_n (q_0 + q_n)}{\sum p_n (q_0 + q_n)}$$

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Interchanging the factors p's and q's, the quantity index is obtained as

$$Q_{0n} = \frac{\sum q_n (p_0 + p_n)}{\sum q_n (p_0 + p_n)}$$

But $$\frac{\sum p_n (q_0 + q_n)}{\sum p_n (q_0 + q_n)} \times \frac{\sum q_n (p_0 + p_n)}{\sum q_n (p_0 + p_n)} \neq \frac{\sum p_n q_n}{\sum p_0 q_0},$$

i.e. (price index) (quantity index) ≠ value index.

Thus the factor-reversal test is not satisfied.

iv. By definition, Fisher' price index is given by

$$P_{0n} = \sqrt{\frac{\sum p_n q_0}{\sum p_0 q_0} \times \frac{\sum p_n q_n}{\sum p_0 q_n}}$$

Fisher's quantity index is obtained by interchanging the factors p's and q's as

$$P_{0n} = \sqrt{\frac{\sum p_n q_0}{\sum p_0 q_0} \times \frac{\sum p_n q_n}{\sum p_0 q_n}}$$

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Therefore $P_{0n} \times Q_{0n}$ = $$\sqrt{\frac{\sum p_n q_0}{\sum p_0 q_0} \times \frac{\sum p_n q_n}{\sum p_0 q_n} \times \frac{\sum q_n p_0}{\sum q_0 p_0} \times \frac{\sum q_n p_n}{\sum q_0 p_n}} = \frac{\sum p_n q_n}{\sum p_0 q_0}$$

i.e. (price index) (quantity index) = value index.

Thus the factor-reversal test is satisfied.

Circular Test

The circular test is an extension of the time reversal test, which is applicable for two years, i.e. the given year and the base year. But in circular test, three or more years are taken into account. This test may be stated thus:

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"If P_{ab} is the price index for the year 'b' with year 'a' as base, P_{bc} is the price index for the year 'c' with year 'b' as base and P_{ca} is the price index for the year 'a' with year 'c' as base, then the product of these index numbers should equal 1. In other words, the circular test requires.

$$P_{ab} \times P_{bc} \times P_{ca} = 1$$

This test is not obeyed by any of the weighted index numbers unless the weights are constant. For the purposes of illustration, we consider the weighted aggregate price index number with fixed weights, say, q0.

Let there be three years denoted by 'a', 'b', and 'c'.

Then the weighted price index numbers are

$$P_{ab} = \frac{\sum p_b q_0}{\sum p_a q_0}, \quad P_{bc} = \frac{\sum p_c q_0}{\sum p_b q_0} \text{ and } P_{ca} = \frac{\sum p_a q_0}{\sum p_c q_0}$$

Multiplying them together, we get

$$P_{ab} \times P_{bc} \times P_{ca} = \frac{\sum p_b q_0}{\sum p_a q_0} \times \frac{\sum p_c q_0}{\sum p_b q_0} \times \frac{\sum p_a q_0}{\sum p_c q_0} = 1.$$

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Which shows that the circular test is satisfied.

Example

Show with the help of the following data that the factor-reversal and time-reversal tests are satisfied by Fisher' Ideal formula for index number construction.

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Commodity	Base Year		Current Year	
	Price (in Rs.)	Quantity (in mds.)	Price (in Rs.)	Quantity (in mds.)
A	6	50	10	56
B	4	100	2	120
C	2	60	6	60
D	10	30	12	24
E	8	40	12	36

Solution

Comm-odity	Price		Quantity		p_0q_0	p_1q_0	p_0q_1	p_1q_1
	p_o	p_1	q_o	q_1				
A	6	10	50	56	300	500	336	560
B	4	2	100	120	400	200	480	240
C	2	6	60	60	120	360	120	360
D	10	12	30	24	300	360	240	288
E	8	12	40	36	320	480	228	432
Total					1440	1900	1464	1880

Now Fisher's price index for the current year (omitting the factor 100) is given by

$$P_{01} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}}$$

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$$= \sqrt{\frac{1900}{1440} \times \frac{1880}{1464}},$$

And Fisher's quantity index for the current year (omitting the factor 100) is given by:

$$Q_{01} = \sqrt{\frac{\sum q_1 p_0}{\sum q_0 p_0} \times \frac{\sum q_1 p_1}{\sum q_0 p_1}} = \sqrt{\frac{1464}{1440} \times \frac{1880}{1900}}$$

$$P_{01} \times Q_{01} = \sqrt{\frac{1900}{1440} \times \frac{1880}{1464}} \times \sqrt{\frac{1464}{1440} \times \frac{1880}{1900}}$$

$$= \sqrt{\frac{1880}{1440} \times \frac{1880}{1400}} = \frac{1880}{1440} = \frac{\sum p_1 q_1}{\sum p_0 q_0}$$ (Value Index).

Hence we see that Fisher's Ideal formula for index numbers satisfies the factor-reversal test.

Again interchanging the time subscripts in Fisher's price index (omitting the factor 100), we get

$$P_{01} = \sqrt{\frac{\sum p_0 q_1}{\sum p_1 q_1} \times \frac{\sum p_0 q_0}{\sum p_1 q_0}}$$

Thus,

$$P_{01} \times P_{10} = \sqrt{\frac{1900}{1440} \times \frac{1880}{1464}} \times \sqrt{\frac{1464}{1880} \times \frac{1440}{1900}} = 1$$

Hence Fisher's Ideal formula for index numbers also satisfies ths time-reversal test.

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Consumer Price Index

- A consumer price index (CPI) is designed to measure the changes in the composite price of a specified "basket" of goods and services during the given period as compared with the base period.
- The quantities consumed or the expenditures incurred on various groups are used as weights for the average retail prices prevailing in the locality concerned during the base and given period.
- A consumer price index is also called a household-budget price index or a retail price index.

Construction of consumer price index

- Scope
- Household Budget Inquiry and Allocation of Weights
- Price Data
- Computation of Index

i. The Aggregate Expenditure

ii. Household Budget Method

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Aggregate Expenditure Method

- The quantities consumed by households in the base year are taken as the weights.
- The quantity represents the aggregate expenditure acquired on the various items in the base year.
- The quantity represents the given year with base year quantities.
- The laspeyres' formula is given by:

$$P_{0n} = \frac{\sum p_n q_0}{\sum p_0 q_0} \times 100$$

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- The given year quantities are not used as weights because they change from year to year.
- It involves expense, labor and time would be needed every year.

Household Budget Method

The method is called Household Budget Method, because the amounts of money spent by the household concerned are obtained from a household consumption survey.

This method is also known as the Weighted Average of Relatives.

$$P_{0n} = \frac{\sum W\left(\frac{p_n}{p_0}\right)}{\sum W} \times 100 = \frac{\sum W \times I}{\sum W}$$

Where W may be $p_0 q_0$ and $I = \frac{p_n}{p_0} \times 100.$

Example

Compute the consumer price index number for 1940 on the basis of 1939 from the following data, using

i. Aggregative Expenditure Method

ii. Household Budget Method.

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Commodity	Quantity Consumed In 1939	Unit Of Price	Price in 1939	Price in 1940
Rice	6 maunds	Rs. per md.	5.75	6.00
Wheat	6 maunds	Rs. per md.	5.00	8.00
Gram	1 maund	Rs. per md.	6.00	9.00
Sugar	1 maund	Rs. per md.	20.00	15.00
Arhar	6 maunds	Rs. per md.	8.00	10.00
Ghee	4 seers	Rs. per md.	2.00	1.50

Solution

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i. Aggregate Expenditure Method:

Commodity	Quantity Consumed In 1939	Unit Of Price	Price in 1939	Price in 1940		
Rice	6 maunds	Rs. per md.	5.75	6.00	34.50	36.00
Wheat	6 maunds	Rs. per md.	5.00	8.00	30.00	48.00
Gram	1 maund	Rs. per md.	6.00	9.00	6.00	9.00
Sugar	1 maund	Rs. per md.	20.00	15.00	20.00	15.00
Arhar	6 maunds	Rs. per md.	8.00	10.00	48.00	60.00
Ghee	4 seers	Rs. per md.	2.00	1.50	8.00	6.00
Total					146.50	174.00

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Aggregate Expenditure Method

Consumer price index for 1940 $= \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100$

$$= \frac{174.00}{146.50} \times 100$$

$$= 118.8.$$

ii. Household Budget Method:

Commodity	Quantity Consumed In 1939	Unit Of Price	Price in 1939	Price in 1940	Weight W	Price relative	
Rice	6 maunds	Rs. per md.	5.75	6.00	34.50	104.30	3598.36
Wheat	6 maunds	Rs. per md.	5.00	8.00	30.00	160.00	4800.00
Gram	1 maund	Rs. per md.	6.00	9.00	6.00	150.00	900.00

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Sugar	1 maund	Rs. per md.	20.00	15.00	20.00	75.00	1500.00
Arhar	6 maunds	Rs. per md.	8.00	10.00	48.00	125.00	6000.00
Ghee	4 seers	Rs. per md.	2.00	1.50	8.00	75.00	600.00
Total					146.50	174.00	17398.36

Consumer price index for 1940

$$= \frac{\sum W \times I}{\sum W}$$

$$= \frac{17398.36}{146.50}$$

= 118.8

Sensitive Price Indicator

- The degree to which the price of a product affects the consumers purchasing behaviors.
- The degree of price sensitivity varies from product to product and from consumer to consumer.
- In economics, price sensitivity is commonly measured using the price elasticity demand.

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Lecture 5

- Drawbacks of Consumer Price Index
- Uses of Index Numbers
- Limitations of Index Numbers
- Practical Questions

Drawbacks of Consumer Price Index

- It is practically difficult to clearly define one category of people from another.
- As the construction of consumer price indices involves the sampling of goods and services, the sampling error and biases may affect indices and render them to suspect.
- In case of certain goods, it is difficult to collect prices actually needed.
- It is also difficult to eliminate the effect of changes in quality and grade of goods and services purch
- ased by households.
- During the course of household budget inquiry, the prices of good and services and their demand may change.
- The consumer price indices cannot be used for comparing the price changes in consumption goods and services in two localities or in two households in the same locality as no two households can be homogeneous.

Uses of Index Numbers

- The price index numbers are used to measure changes in a particular group of prices and help us in comparing the movement in prices of one commodity with another.
- Index numbers of industrial production provide a measure of change in the level of industrial production in a country.
- The quantity index numbers show the rise or fall in the volume of production, volume of exports and imports, etc.

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- The import and export price indices are used to measure the changes in the terms of trade of a country.
- Index numbers are also used to forecast business conditions of a country and to discover seasonal fluctuations and business cycles.
- The consumer price indices indicate the movements in retail price of consumption goods and services.
- Index numbers are also used to measure enrolment changes, intelligence quotients and the performance of students.

Limitations of Index Numbers

- It is not practicable to price all the goods and services as well as to take into account all changes in quantity or product.
- Since the construction of almost all index numbers is based on sampling of some kind, therefore the sampling errors creep into their calculations.
- In price indices the choice of a normal period is difficult as few periods can be regarded as normal for all segments of the economy.
- The results obtained by different methods of construction may not quite agree.
- Comparisons of changes in variables over long periods are not reliable.
- All index numbers are not suitable for all purposes.

Example

The prices of four commodities quoted at Multan for May 1970, April 1971 and May 1971 are given below:

Commodity	May 1970	April 1971	May 1971
Wheat	17.50	18.37	17.58

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Barley	14.58	14.58	16.50
Jawar	14.67	13.94	15.25
Bajra	17.50	13.75	13.42

Compute price index numbers for April 1971 and May 1971 with May 1970 as base, using (i) simple aggregative method, (ii) simple average of price relatives and (iii) geometric mean of price relatives.

Solution

1. Simple aggregative method:

$$P_{0n} = \frac{\sum p_n}{\sum p_0} \times 100$$

i. Simple aggregative method for April 1971:

$$P_{01} = \frac{18.37 + 14.58 + 13.94 + 13.75}{17.50 + 14.58 + 14.67 + 17.50} \times 100 = 94.38$$

ii. Simple aggregative method for May 1971:

$$P_{02} = \frac{17.58 + 16.50 + 15.25 + 13.42}{17.50 + 14.58 + 14.67 + 17.50} \times 100 = 97.67$$

2. Simple average of price relatives:

	Price relatives	**Sumof**	**Index No.**

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Year	Wheat	Barley	Jawar	Bajra	price relatives	As average of relatives
May 1970	100	100	100	100	400	100
April 1971	105.0	100	95.0	78.6	378.6	94.65
May 1971	100.5	113.2	104.0	76.7	394.4	98.60

3. Geometric mean of price relatives:

	Price relatives						Indices
Year	**Wheat**	**Barley**	**Jawar**	**Bajra**			
May 1970	100	100	100	100	8.0000	2.0000	100
Log	2.0000	2.0000	2.0000	2.0000			
April 1971	105.0	100	95.0	78.6	7.8943	1.9738	94.14
Log	2.0212	2.0000	1.9777	1.8954			

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May 1971	100.5	113.2	104.0	76.7	7.9578	1.9894	97.59
Log	2.0021	2.0539	2.0170	1.8848			

Example

Construct the Chain Indices for the following data for price relatives for 1941 to 1944.

Year	Sugar	Gur	Tea	Coffee
1941	98	75	82	99
1942	100	82	74	100
1943	114	83	78	104
1944	109	84	89	95

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Year	Link Relatives				TotalofL.R	Average	Chain Indices
	Sugar	Gur	Tea	Coffee			
1941	98	75	82	99	354	88.5	88.5
1942	102	109	90	101	402	100.5	88.9
1943	114	101	105	104	424	106.0	94.2

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1944	96	101	114	91	402	100.5	94.7

Example

Given the following, construct Fisher's Ideal index number for

i. 1964, taking 1957 as base year.

ii. 1957, taking 1964 as base year.

Commodity	**1957**		**1964**	
	Price	**Quantity**	**Price**	**Quantity**
Rice	9.3	100	4.5	99
Wheat	6.4	11	3.7	19
Jawar	5.1	5	2.7	3

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1. Fisher's Ideal index number for

i. 1964, taking 1957 as base year.

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Commodity	1957 Base Year		1964 Current Year		p_0q_0	p_1q_0	p_0q_1	p_1q_1
	Price	Quantity	Price	Quantity				
Rice	9.3	100	4.5	90	930.0	450.0	837.0	405.0
Wheat	6.4	11	3.7	19	70.4	40.7	121.6	70.3
Jawar	5.1	5	2.7	3	25.5	13.5	15.3	8.1

$$\sqrt{\frac{\sum p_1q_0}{\sum p_0q_0} \times \frac{\sum p_1q_1}{\sum p_0q_1}} \times 100$$

$$\sqrt{\frac{504.2}{1025.5} \times \frac{483.5}{973.9}} \times 100 = 49.4$$

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1. Fisher's Ideal index number for

ii. 1957, taking 1964 as base year.

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Commodity	1957 Current Year		1964 Base Year		p_0q_0	p_1q_0	p_0q_1	p_1q_1
	Price	Quantity	Price	Quantity				
Rice	9.3	100	4.5	90	405	837	450	930
Wheat	6.4	11	3.7	19	70.3	121.6	40.7	70.4
Jawar	5.1	5	2.7	3	8.1	15.3	13.5	25.5
					483.4	973.9	504.2	1025.9

$$\sqrt{\frac{\sum p_1q_0}{\sum p_0q_0} \times \frac{\sum p_1q_1}{\sum p_0q_1}} \times 100$$

$$\sqrt{\frac{973.9}{483.4} \times \frac{1025.9}{504.2}} \times 100 = 202.5$$

Example

The prices and quantities of three commodities during 1955 and 1965 are given below:

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Commodity	1955		1965	
	Price	Quantity	Price	Quantity
A	10	501	12	600
B	40	100	38	194
C	50	76	40	56

Using 1955 as the base period and base period quantities as the weights, compute the weighted aggregative price index and the weighted average of relative price index for 1965.

1. Weighted Aggregate Price Index for 1965:

Commodity	1955		1965		p_1q_0	p_0q_0
	Price	Quantity	Price	Quantity		
A	10	501	12	600	6012	5010
B	40	100	38	194	3800	4000
C	50	76	40	56	3040	3800

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$$P_{0n} = \frac{\sum p_n q_0}{\sum p_0 q_0} \times 100$$

$$P_{0n} = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100$$

$$P_{0n} = \frac{12852}{12810} \times 100 = 100.33$$

2. Weighted Average of Relative Price Index for 1965:

Commodity	1955		1965		$\frac{p_1}{p_0}$	$\left(\frac{p_1}{p_0}\right) p_0 q_0$
	Price	Quantity	Price	Quantity		
A	10	501	12	600	1.20	6012
B	40	100	38	194	0.95	3800
C	50	76	40	56	0.80	3040

$$P_{0n} = \frac{\sum (p_n / q_0) p_0 q_0}{\sum p_0 q_0} \times 100$$

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$$P_{0n} = \frac{\sum \left(p_1 / q_0\right) p_0 q_0}{\sum p_0 q_0} \times 100$$

$$P_{0n} = \frac{12852}{12810} \times 100 = 100.33$$

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Lecture 6

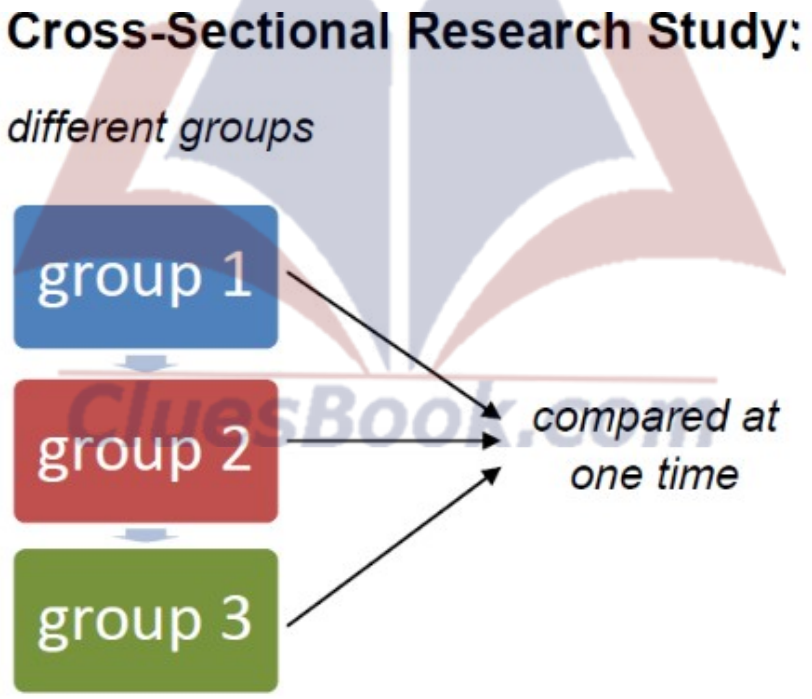
Time series and Data Analysis

Structures of Data

- Cross-Sectional Data
- Time series Data
- Panel Data

Cross Sectional Data

These are data from units observed at the same time or in the same time period. The data may be single observations from a sample survey or from all units in a population.

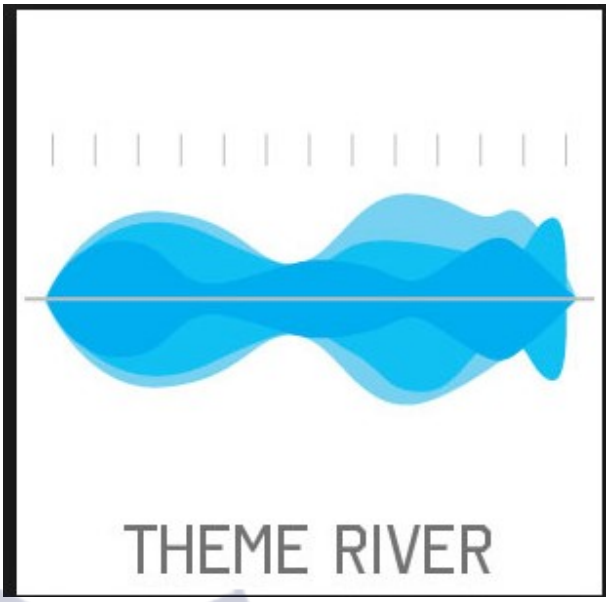


Time Series Data

These are data from a unit (or a group of units) observed in several successive periods.

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Panel Data

Panel data, also called longitudinal data or cross sectional time series data. Panel data are data where multiple cases (people, firms, countries etc).

Panel data were observed at two or more time periods. An example is the National Longitudinal Survey of Youth, where a nationally representative sample of young people were each surveyed repeatedly over multiple years.

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Objective of Time Series Analysis

- Compact description of data
- Interpretation
- Forecasting
- Control
- Hypothesis testing
- Simulation

Lecture Content

- Time Series
- Historigram
- Component of Time Series

Time Series

- A time series consists of numerical data collected, observed or recorded at more or less regular intervals of time each hour, day, month, quarter or year.
- It is set of data in which observations are arranged in a chronological order.

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Example

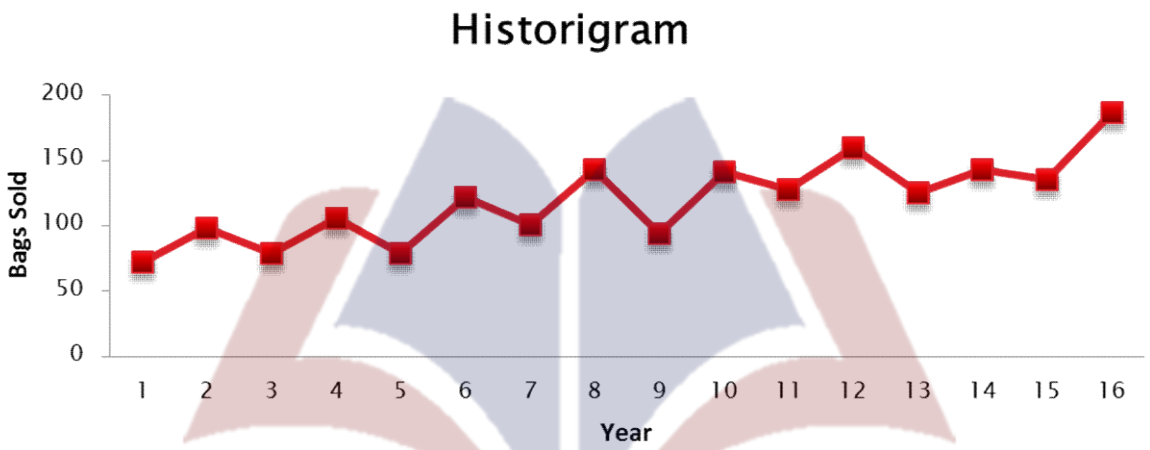
- Hourly temperature recorded at a area.
- Weekly prices of wheat in Lahore.
- Monthly consumption of electricity in a certain town.
- Monthly total of passengers carried by rail.
- Quarterly sales of a certain fertilizer.
- Annual rainfall at Karachi for a number of years.
- Enrolment of students in a college over a number of years.

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Graph of Time Series

- A curve showing changes in the value of one or more items from one period to next is known as the graph of a time series. This curve is called a *Historigram*.
- Historigram displays the variations in time series dealing with prices, production, imports, population, etc.
- To construct a *historigram,* time is taken along X-axis and the values of the variable along Y-axis. Points are plotted and then connected the points by the straight line.



Component of Time Series

- Secular Trend
- Seasonal Variation
- Cyclic Variation
- Random Variation

Secular Trend

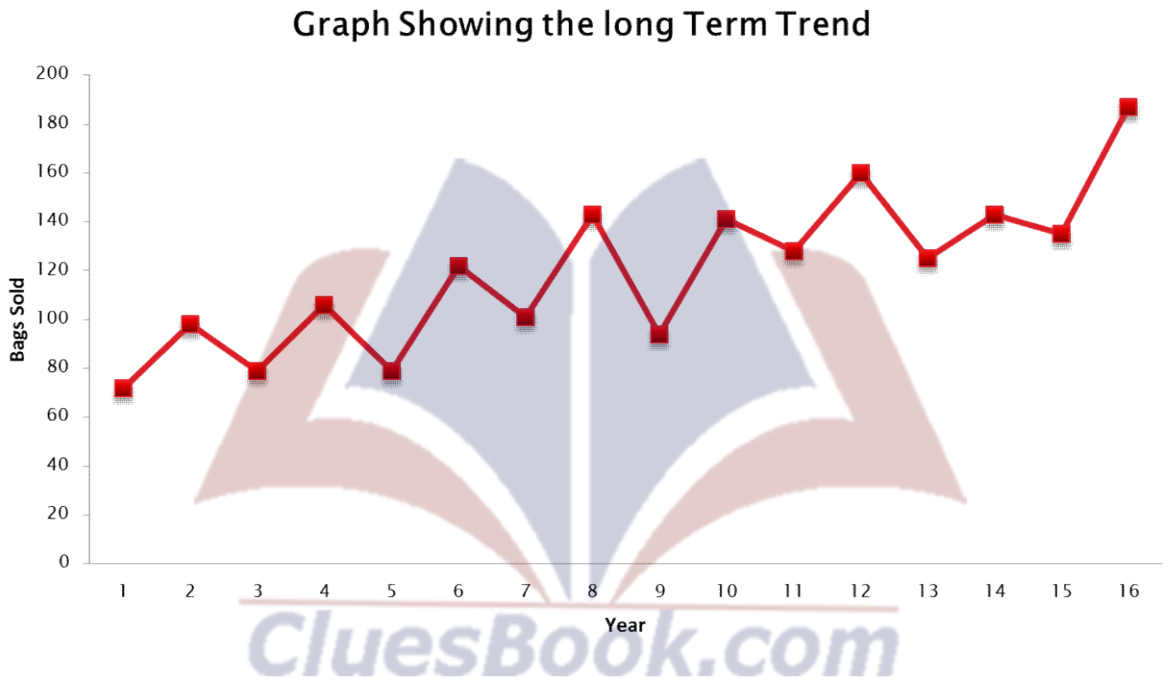
- A secular trend (T) is a long-term movement that persists (Continues) for many years and indicates the general direction of the change of observed values.
- It refers to a smooth, broad movement of a time series in a same direction, showing a gradual rise or fall within the data.

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- The secular trend generally dominates other variations in the long run and covers a fairly long period of time, not less than 10 years.
- The long term trend is a peculiar characteristics of most of the economic variables such as sales, prices, industrial production, capital information, etc.
- It also helps in business planning.

Graph of Secular Trend



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Seasonal Variations

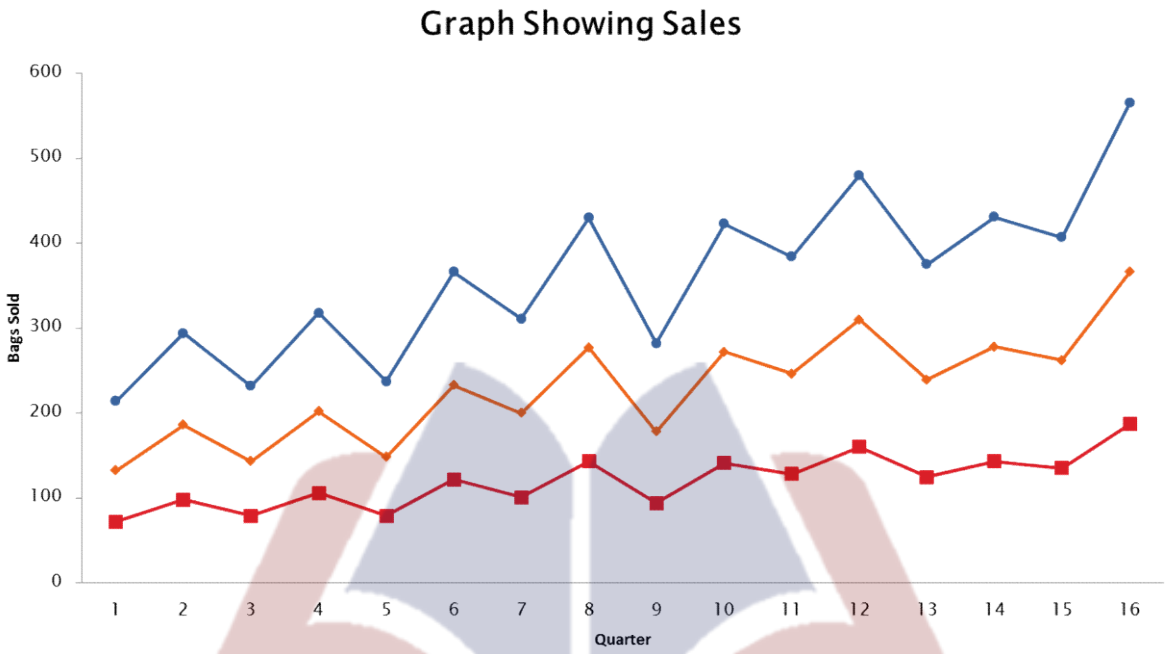
- The seasonal variations (S), which are mainly caused by the change in seasons, are short-term movements occurring in a periodic manner.
- The seasonal variations are repeated with more or less the same intensity within a specific period of one year or shorter .
- The main causes for seasonal variations are the weather conditions, the religious festivals and the social customs.
- The concept of seasonal variation is occurring within a day, a week, a month, a quarter and so forth.

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- Examples of such variations are the daily variations in temperature or the monthly variations in Bank deposits.

Graph of Seasonal Trend



Cyclical Fluctuations

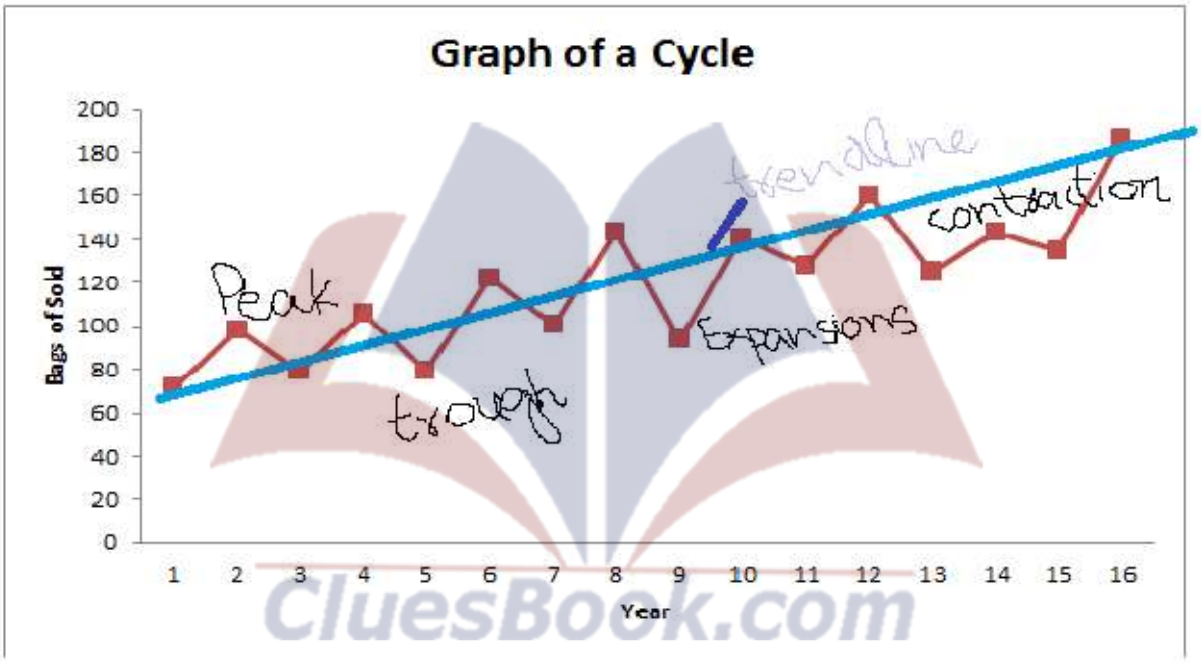
- Cyclical fluctuations are the long-period oscillations about the long-term trend, which trend to occur in a more or less regular pattern over a period of certain number of years.
- It is also called business cycles which represent alternating period of prosperity and depression.
- A cycle, as it is known, is said to be completed when beginning with a peak (a peak is a value which is greater than the two neighboring values), the declining curve reaches a low point, called a trough (a trough is a value which is lower than the two neighboring values) and then rising again reaches the next peak.
- The period from peak to peak or from trough to trough, is referred to as the duration of a cycle.
- Cycles have a duration of anywhere from two to ten years or even a longer period.

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- In general, a complete business cycle has the following four phases:
- Period of prosperity
- Period of contraction
- Period of recession or depression
- Period of recovery or expansion

Graph of Cyclical fluctuation



Random Variations

- These variations are irregular and unsystematic in nature.
- These variations occur in a completely unpredictable manner as they are caused by some unusual events such as floods, droughts, strikes, fires, earthquakes, wars, locusts, political events, etc.
- These variations are also known as the accidental, residual or erratic variations.
- It is difficult to make a study of such non-recurring variations, though they can be easily identified.

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Practice Questions

Associate the following phenomena of business with the components of time series they belong to:

- The prosperity in a business.
- The production of sugar recorded for 1956,1957,….,1962.
- The weekly statement of the sale of pens.
- The festival sale.
- The fire in a factory.

With which characteristic movement of a time series would you mainly associate each of the following?

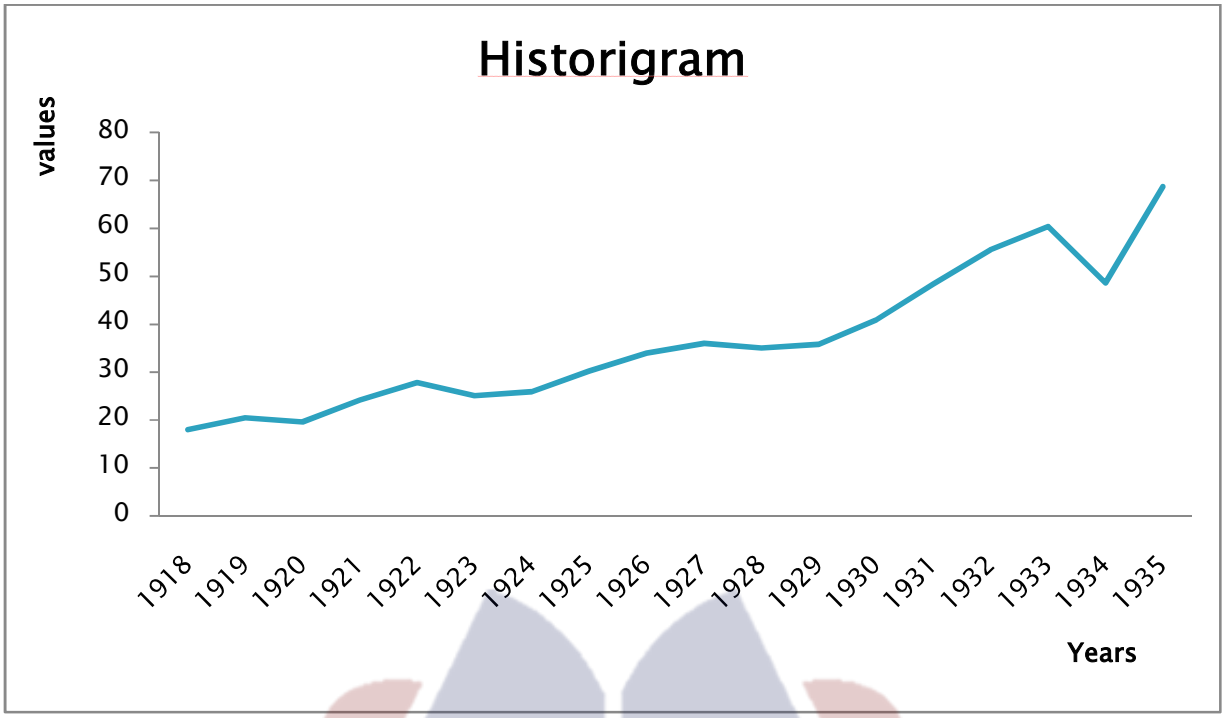
- A recession
- An increase in employment during summer months
- The decline in the death rate due to advance in science
- A steel strike
- A continually increasing demand for smaller automobiles.

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How to draw graphs on Excel

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Lecture 7

- Time Series Decomposition
- Analysis of Secular Trend
- Moving Average

Time Series Decomposition

- A time series analysis is mainly concerned with the decomposition of the observed series data into its components so as to estimate their separate effects.
- To do this, we must make assumptions about the relationship existing among the various components. Such as:

I. Multiplicative Model

II. Additive Model

Multiplicative Model

- In multiplicative model, we assume that each observed value at any time can be determined by the product of the measures of all four components T, C, S and I. Symbolically,

$$Y_t = T \times S \times C \times I = TSCI$$

- Where observed values and T-values are stated in original units but the other components S, C and I are expressed in percentages.

Additive Model

- In additive model, each value that we observed, is thought of as being a sum of all the four components.

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$$Y_t = T + S + C + I$$

- Where the components T, S, C and I are assumed to be mutually independent.
- Before analyzing a time series, it is often desirable to adjust the data for calendar variations, for holidays, for price changes and so forth.

Analyzing the Secular Trend

- The analysis of a trend component involves its measurement and/or elimination from an observed time series data.
- To measure a trend which can be represented as a straight line or some type of smooth curve, the following methods are used.

i. Method of freehand curve

ii. Method of Semi-averages

iii. Method of moving averages

iv. Method of least-squares.

Method of Freehand Curve

- Plot the given data on a graph paper and join the plotted points by segments of straight line.
- Observe the up and down movements on the graph and draw a smooth curve or a straight line freehand passing though the plotted points that show the direction of change.
- This line may also be drawn by a transparent ruler.
- This method is simple and quick.

Disadvantages of Method of Freehand Curve

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The freehand curve is a rough and subjective method. As the drawing of trends depends on individual judgment and experience, different persons will draw the graphical trend at different positions with different slopes.

Method of Semi-Averages

- Divide the values in the series into two equal parts, including the middle value in each half or omitting it altogether when the number of values is odd.
- Find the average of the values in each part and place the average values against the respective midpoints of the two parts.
- Plot these two average values on the graph of the original values, draw a straight line connecting the two points and extend the line to cover the whole series.
- This method is simple and quick. It gives an entirely objective result when the trend is a straight line.

Disadvantages of Method of Semi-Averages

- The arithmetic mean is used to average the observed values is greatly affected by abnormally small or large values.
- The method is only suitable when the trend is linear or nearly linear.

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Example

Compute and insert on graphs the semi-average trends for the following series:

a) Annual profits in thousands of rupees in a certain business.

Year	1973	1974	1975	1976	1977	1978	1979	1980
Profit	85	97	100	90	83	105	112	120

b) Indoor patients (‘000) treated in Hospitals in the Punjab.

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	1967	1968	1969	1970	1971	1972	1973	1974	1975
Patients	276	270	260	286	302	321	351	348	346

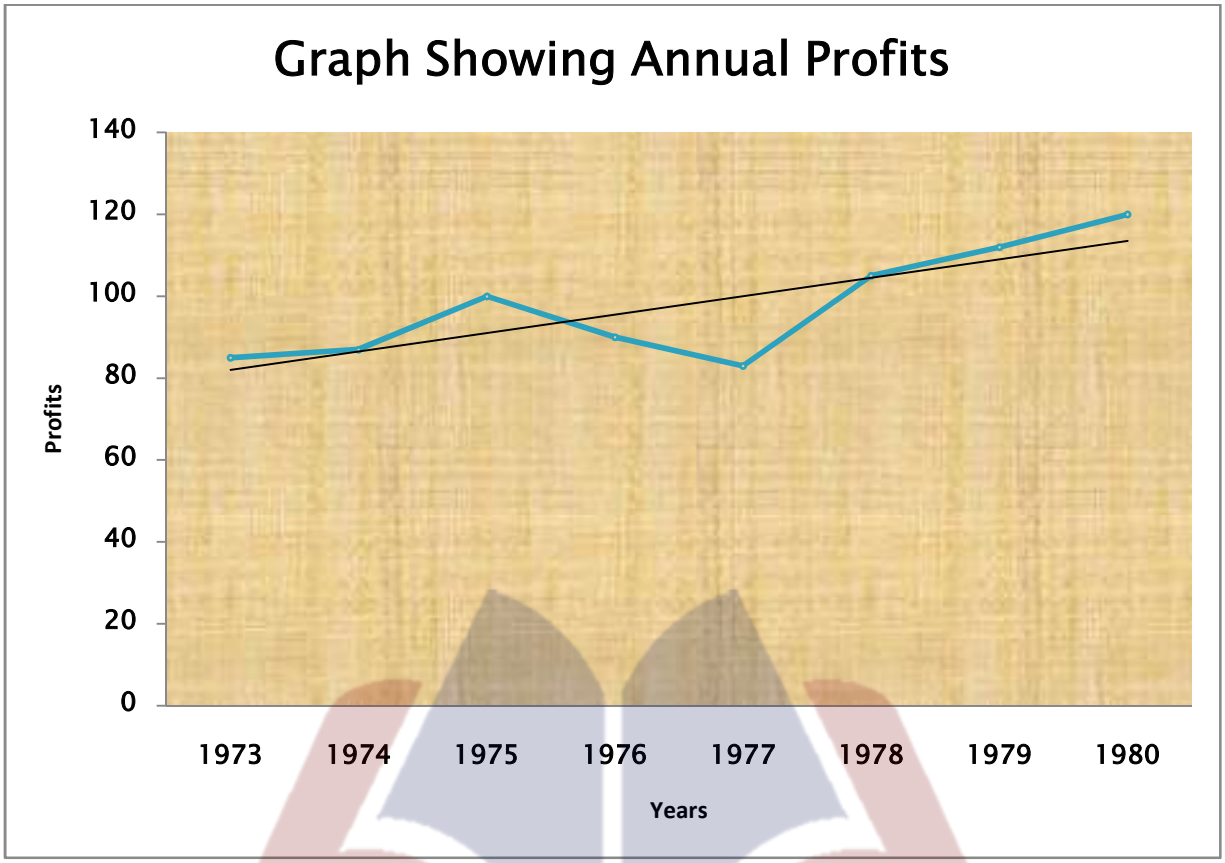
a)

Year	Profit	Total	Averages
1973	85		
1974	97	372	93
1975	100		
1976	90		
1977	83		
1978	105	420	105
1979	112		
1980	120		

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b)

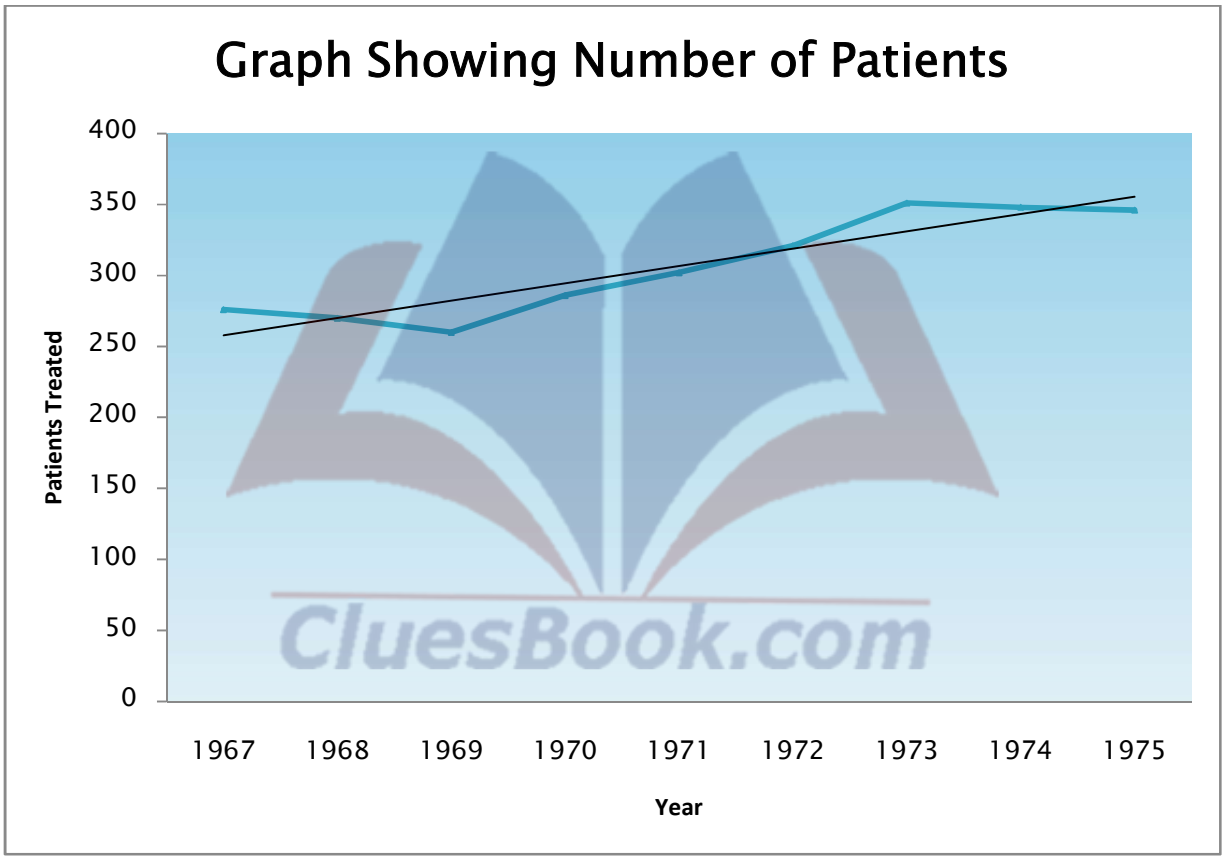
Year	Patients	Total	Averages
1967	276		
1968	270		
1969	260	1092	273
1970	286		
1971	302		
1972	321		

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1973	351	1366	341.5
1974	348		
1975	346		



Example

Plot the following data showing average wages in rupees of some workers during 1970-1981. Compute the method of

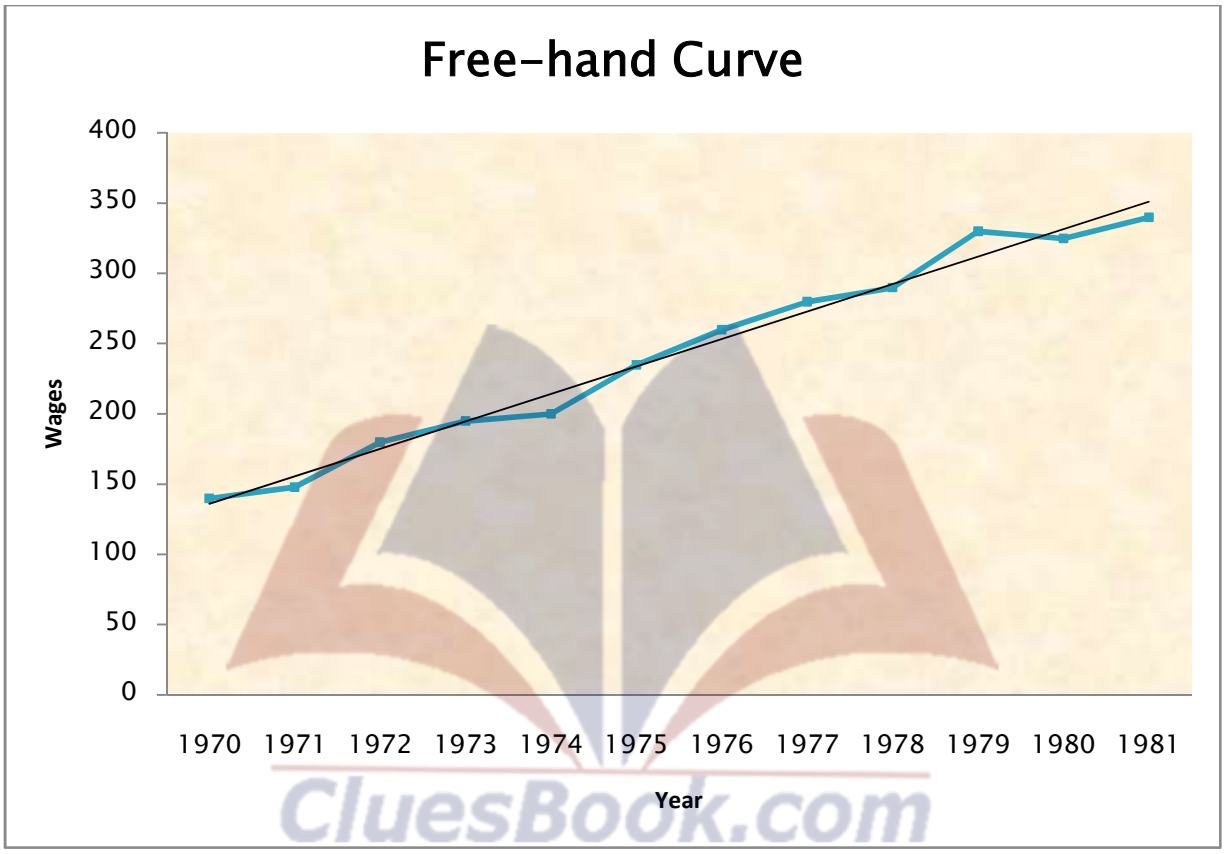
i. Free-hand curve

ii. Semi-averages

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Year	1970	1971	1972	1973	1974	1974	1975	1976	1977	1978	1979	1980
Wages	140	148	180	195	200	235	260	280	290	330	325	340



ii. Semi-averages

Year	Wages	Total	Averages
1970	140	1098	183
1971	148		
1972	180		

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1973	195		
1974	200		
1975	235		
1976	260		
1977	280		
1978	290	1825	304.17
1979	330		
1980	325		
1981	340		

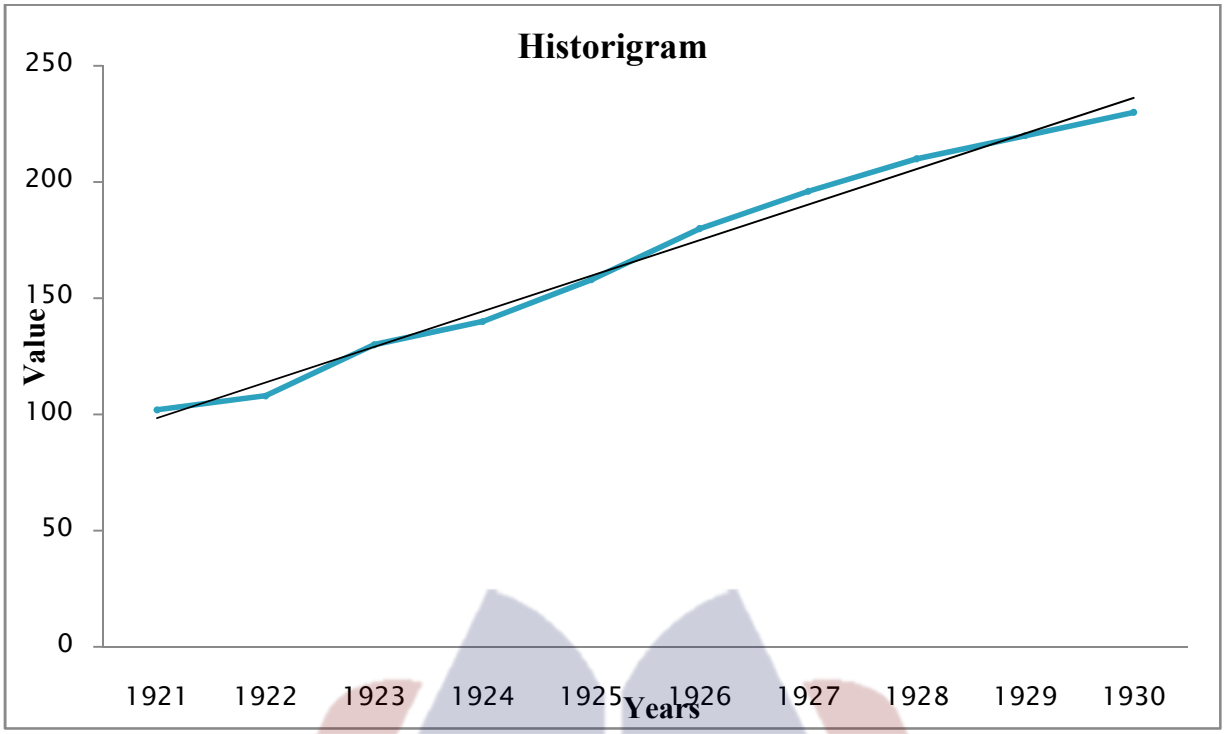
Example

Compute and insert on graphs the semi-average trends for the following series:

Year	1921	1922	1923	1924	1925	1926	1927	1928	1929	1930
Value	102	108	130	140	158	180	196	210	220	230

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ii) Semi average

Year	Wages	Total	Averages
1921	140		
1922	148		
1923	180	863	172.6
1924	195		
1925	200		
1926	235		
1927	260		

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1928	280	1395	279
1929	290		
1930	330		

Example

Plot the following data showing average wages in rupees of some workers during 1970-1981 and find the trend by the method of

i. Free-hand curve

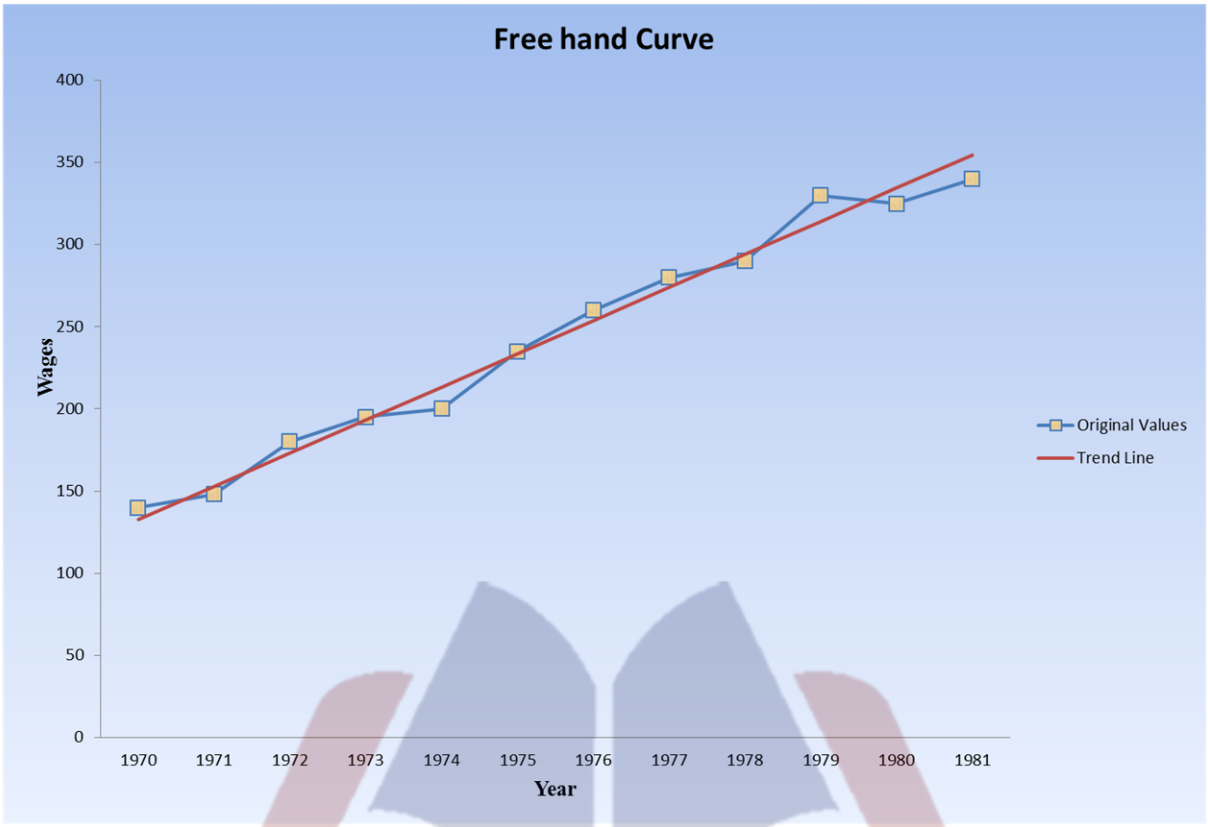
ii. Semi-averages

Year	1970	1971	1972	1973	1974	1974	1975	1976	1977	1978	1979	1980
Wages	140	148	180	195	200	235	260	280	290	330	325	340

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ii) Semi averages

Year	Wages	Total	Averages	Trend Values
1970	140			132.5
1971	148			152.7
1972	180	1098	183	172.9
1973	195			193.1
1974	200			213.3
1975	235			233.5

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1976	260	1825	304	253.7
1977	280			273.9
1978	290			294.1
1979	330			314.3
1980	325			334.5
1981	340			354.7

Method of Moving Average

- The k-period moving averages are defined as the averages calculated by using the k consecutive values of the observed series, then repeating the operation by dropping one value at the beginning and including the first value after the preceding total, and so on, moving on one value to calculate each successive average.
- Each k-period moving average is placed against the middle of its time period.
- The average will directly to the observed value in the series when k is odd and when k is even, the moving average will be placed at points which will be located between two periods.
- For the purposes of illustration, let us first choose k to be odd, say k=3 years, then we compute the 3-year moving averages as

$$a_1 = \frac{1}{3}(Y_1 + Y_2 + Y_3)$$

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$$a_2 = \frac{1}{3}(Y_2 + Y_3 + Y_4)$$

$$a_3 = \frac{1}{3}(Y_3 + Y_4 + Y_5)$$

and so on. The averages so obtained are placed opposite the middle year of each group, i.e. opposite the 2nd year, 3rd year, 4th year and so on.

A 4-year centered moving average

$$a_1^{/} = \frac{1}{2}\left[\frac{1}{4}(Y_1 + Y_2 + Y_3 + Y_4) + \frac{1}{4}(Y_2 + Y_3 + Y_4 + Y_5)\right]$$

$$= \frac{1}{8}[Y_1 + 2Y_2 + 2Y_3 + 2Y_4 + Y_5]$$

That is, a 4-year centered moving average. These averages are plotted on the graph of the original values and the line connecting these points is the moving average trend.

Advantages of Moving Average

- The method of moving averages is easy and simple.
- The moving averages of appropriate period estimate the combined effects of trend and cyclical components and give a smooth version of the series by removing the seasonal and other effects.

Disadvantages of Moving Average

- The moving average trend does not provide values at the end or the beginning of the original series by half the period.
- The simple moving average are affected by large Y-values. It may be reduced by using the geometric mean.

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- In the absence of an appropriate period, moving averages may make the resulting series more cyclical than the observed series.
- The method does not provide the mathematical expression for the trend.

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Lecture 8

- Analysis of Moving Average
- Method of Least Square

Example

Compute "3-year moving averages" for the following data and show the moving average trends on the graph.

Year	1918	1919	1920	1921	1922	1923	1924	1925	1926	1927	1928	1929	1930
Value	18.0	20.5	19.6	24.2	27.8	25.1	25.9	30.2	34.0	36.0	35.0	35.8	40.9

3-year Moving average

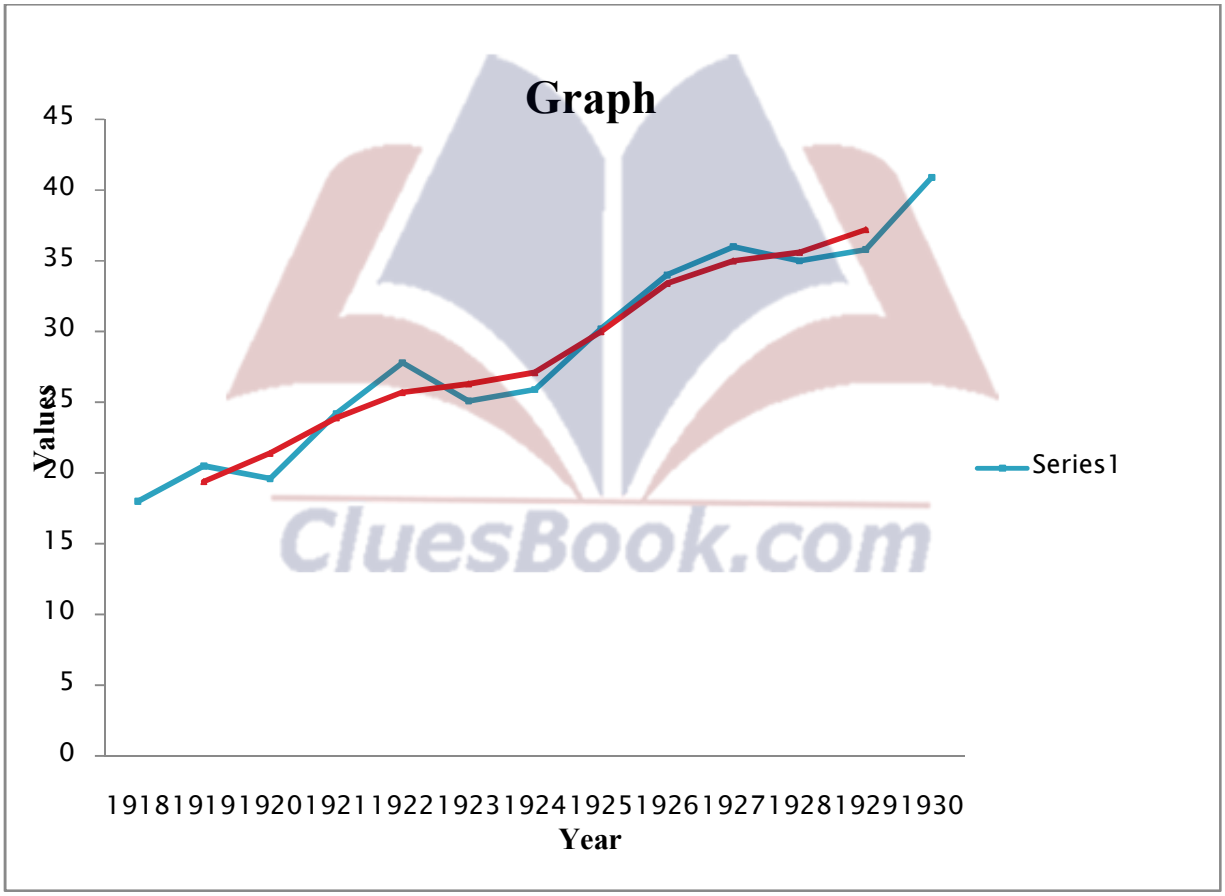
Year	Values	Total	Average
1918	18.0		
1919	20.5	58.1	19.4
1920	19.6	64.3	21.4
1921	24.2	71.6	23.9
1922	27.8	77.1	25.7
1923	25.1	78.8	26.3
1924	25.9	81.2	27.1
1925	30.2	90.1	30.0

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1926	34.0	100.2	33.4
1927	36.0	105.0	35.0
1928	35.0	106.8	35.6
1929	35.8	111.7	37.2



Example

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Compute (i) 5-year moving averages and (ii) 7-year moving averages for the following data and show the moving average trends on the graph.

Year	1918	1919	1920	1921	1922	1923	1924	1925	1926	1927	1928	1929	1930
Value	18.0	20.5	19.6	24.2	27.8	25.1	25.9	30.2	34.0	36.0	35.0	35.8	40.9

5-year Moving Average

Year	Values	Total	Average
1918	18.0		
1919	20.5		
1920	19.6	110.1	22.0
1921	24.2	117.2	23.4
1922	27.8	122.6	24.5
1923	25.1	133.2	26.6
1924	25.9	143.0	28.6
1925	30.2	151.2	30.2
1926	34.0	161.1	32.2
1927	36.0	171.0	34.2

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1928	35.0	181.7	36.3
1929	35.8		
1930	40.9		

7-year Moving average

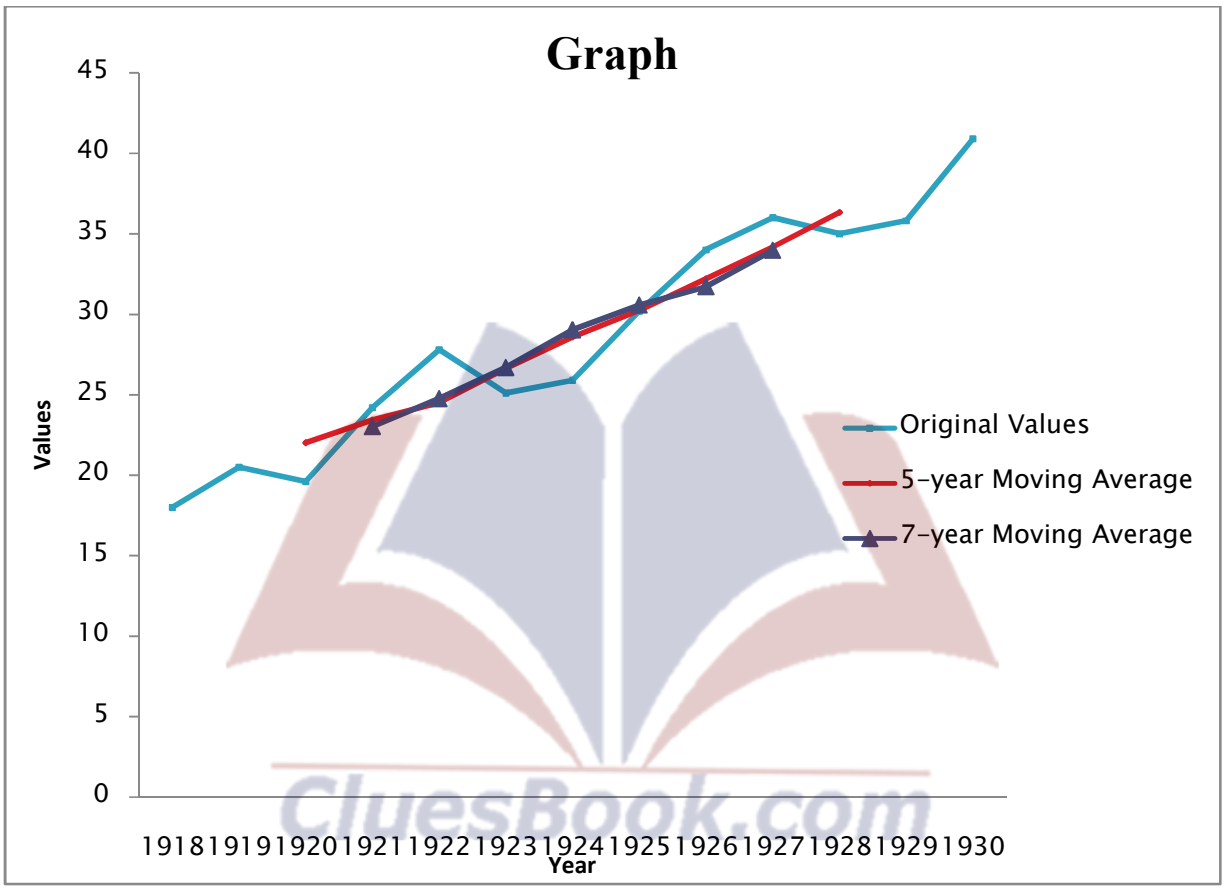
Year	Values	Total	Average
1918	18.0		
1919	20.5		
1920	19.6		
1921	24.2	161.1	23.0
1922	27.8	173.3	24.8
1923	25.1	186.8	26.7
1924	25.9	203.2	29.0
1925	30.2	214.0	30.6
1926	34.0	222.2	31.7
1927	36.0	237.8	34.0
1928	35.0		

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1929	35.8		
1930	40.9		



Example

Determine a line in the following series of values by the use of a simple average of seven consecutive terms.

Year	1924	1925	1926	1927	1928	1929	1930	1931	1932	1933	1934	1935	1936	1937
Index	187	161	149	142	125	129	133	127	130	129	129	130	136	152

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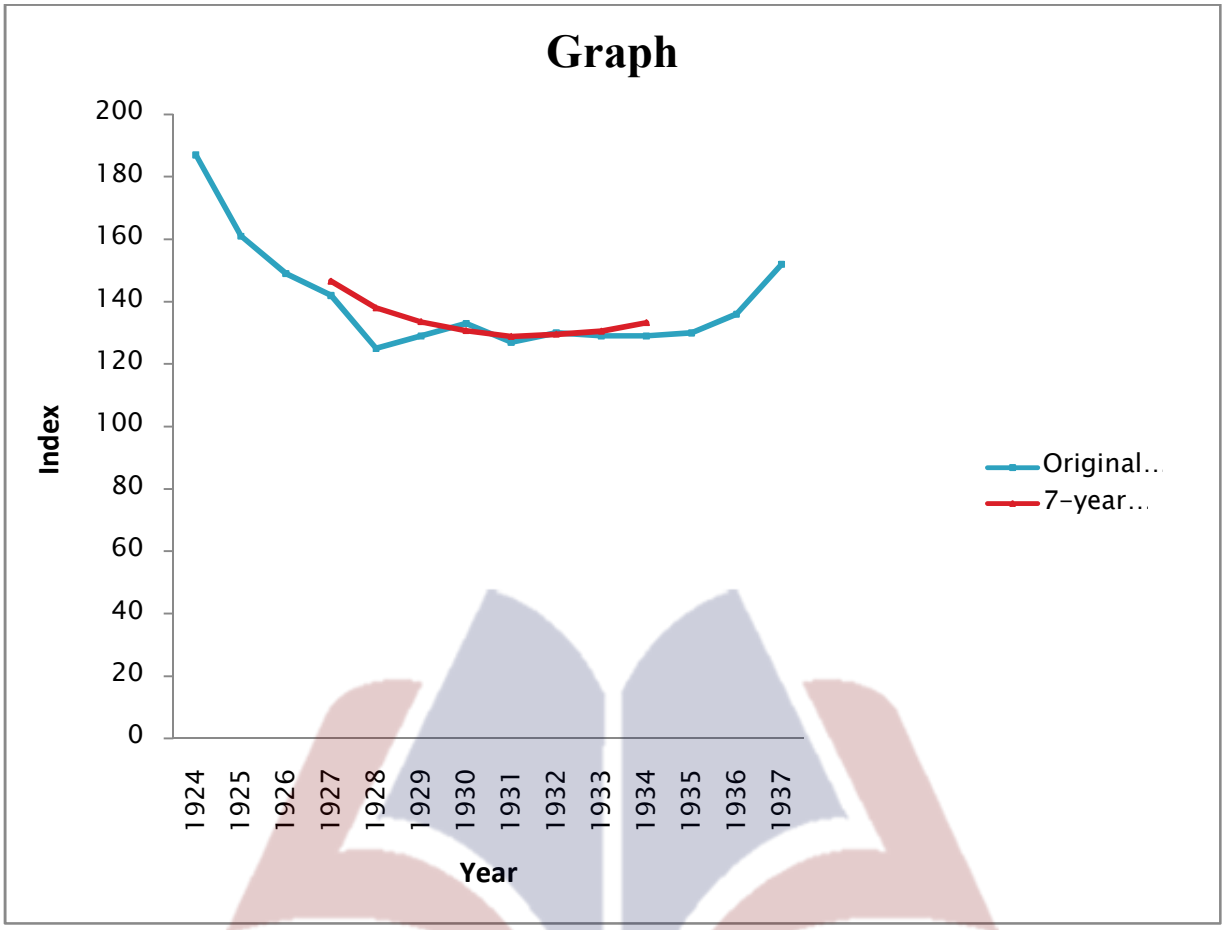
7-year Moving Average

Year	Values	Total	Average
1924	187		
1925	161		
1926	149		
1927	142	1026	146.6
1928	125	966	138.0
1929	129	935	133.6
1930	133	915	130.7
1931	127	902	128.9
1932	130	907	129.6
1933	129	914	130.6
1934	129	933	133.3
1935	130		
1936	136		
1937	152		

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Example

Find out and plot the nine-year moving average for the following series:

8, 7, 5, 2, 4, 9, 10, 9, 8, 6, 4, 7, 11, 13, 11, 9, 8, 5, 10, 13, 15, 12, 10, 8, 6, 11, 12, 16.

Year	Value	9-Year Moving	
		Total	Average
1	8		
2	7	62	6.9
3	5	60	6.7
4	2	57	6.3

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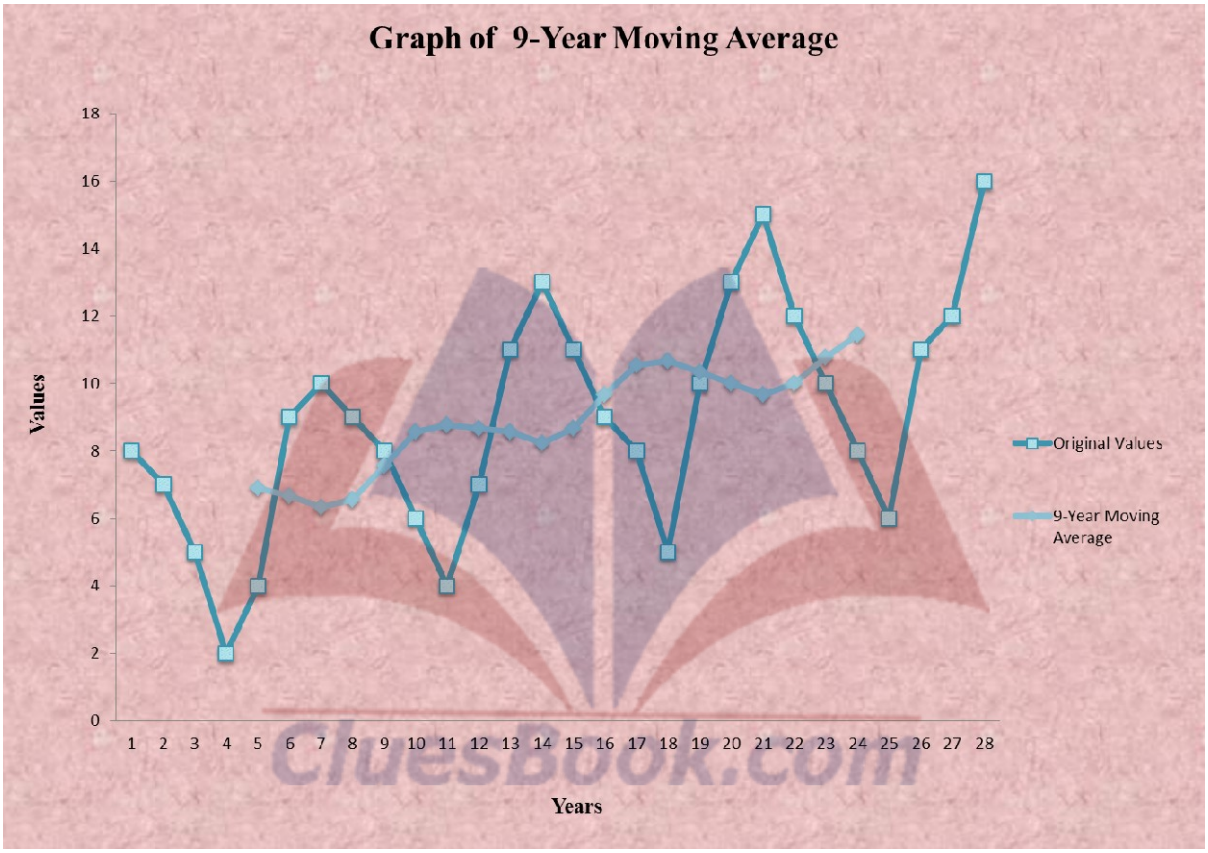
5	4	59	6.6
6	9	68	7.6
7	10	77	8.6
8	9	79	8.8
9	8	78	8.7
10	6	77	8.6
11	4	74	8.2
12	7	78	8.7
13	11	87	9.7
14	13	95	10.6
15	11	96	10.7
16	9	93	10.3
17	8	90	10.0
18	5	87	9.7
19	10	90	10.0
20	13	97	10.8
21	15	103	11.4
22	12		
23	10		
24	8		
25	6		

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26	11		
27	12		
28	16		



Example

Compute 4-month centered moving average from the following:

23, 26, 28, 30, 31, 35, 37, 32, 34, 38.

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Month	Values	4-Month Moving Total	4-Month Centered Moving Totals	4-Month Centered Moving Averages
1	23	107	222	27.75
2	26	115	239	29.88
3	28	124	257	32.12
4	30	133	268	33.50
5	31	135	273	34.12
6	35	138	279	34.88
7	37	141		
8	32			
9	34			
10	38			

Example

The estimated number of visitors (’00s) at a holiday resort were as follows:

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Year	1971	1972	1973	1974	1975	1976	1977	1978	1979	1978
Visitor	31	49	74	62	65	73	70	84	86	79

Calculate the 2-year centered moving average for the above following table.

Year	Visitors	2-year Moving totals	2-year centered moving totals	2-year centered moving average
1971	31	80	203	50.75
1972	49	123	259	64.75
1973	74	136	263	65.75
1974	62	127	265	66.25
1975	65	138	281	70.25
1976	73	143	297	74.25
1977	70	154	324	81.00
1978	84	170	335	83.75
1979	86	165		
1980	79			

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Method of Least Squares

- A trend line, which can be described by a mathematical equation of the form of a straight line, a parabola, or an exponential, can be measured using the method of least squares, by letting time to be the independent variable.
- Let us suppose that the long-term trend appears to be linear. Then the equation of the least squares linear trend would be

$$\hat{Y}_i = a + bt$$

the values of a and b, the two constants, are determined by solving the following two normal equations simultaneously:

$$\sum Y = na + b\sum t,$$

$$\sum ty = a\sum t + b\sum t^2$$

- The trend values are computed from this equation by substituting the values of t and are plotted on the graph of original values.
- To simplify computations, the time variable (t) is coded by taking the time deviations from the middle point of the periods and the coded time period is denoted by X.
- When the number of years is odd, the middle year is taken as 0 or its origin.
- When the number of years is even, X=0 is taken at the middle of two middle years.

Odd number of years		Even number of years	
Year	Coded year	Year	Coded year
1978	-1	1978	-1

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1979	0	1979	-0.5
1980	1	1980	0.5
		1981	1

- Sometimes, the numbers 1, 2, 3, …, may also be assigned to X. The coded form of the linear trend would be $Y_t = a + bX$ and the two-normal equations would reduce to

$$\sum Y = na \quad \text{and} \quad \sum XY = b\sum X^2$$

- When the long-term trend of a time series graph appears to be a curve, we fit a parabola curve of the second or third degree or a curve of some other type such as exponential, modified exponential, Gompertz, logistic, etc. It should be kept in mind that a trend has the smaller $\sum (Y - Y_t)^2$, is the better fit.

Advantages of Least Squares Method

- The method of least square is mathematically sound and is therefore useful for prediction purposes.
- It is also an objective method.

Disadvantages of Least Squares Method

- It is sometime difficult to choose a proper mathematical equation.
- In case of non-linear trends, the method may involve heavy calculations.

Example

For the following time series, determine the trend line by using the method of least squares for fitting a straight line. Also plot the actual values and trend values on the graph.

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Year	1970	1971	1972	1973	1974	1975	1976	1977	1978
Value	12	23	37	48	41	37	48	61	70

Two Normal Equations:

$$a = \frac{\sum Y}{n}$$

$$b = \frac{\sum XY}{\sum X^2}$$

Year (t)	Coded year (X)	Y	XY		Trend
1970	-4	12	-48	16	18.09
1971	-3	23	-69	9	24.04
1972	-2	37	-74	4	29.99
1973	-1	48	-48	1	35.94
1974	0	41	0	0	41.89
1975	1	37	37	1	47.84
1976	2	48	96	4	53.79

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1977	3	61	183	9	59.74
1978	4	70	280	16	65.69
Total	0	377	357	60	377.01

$$a = \frac{\sum Y}{n} = \frac{377}{9} = 41.89$$

$$b = \frac{\sum XY}{\sum X^2} = \frac{357}{60} = 5.95$$

$$Y_t = 41.89 + 5.95X$$

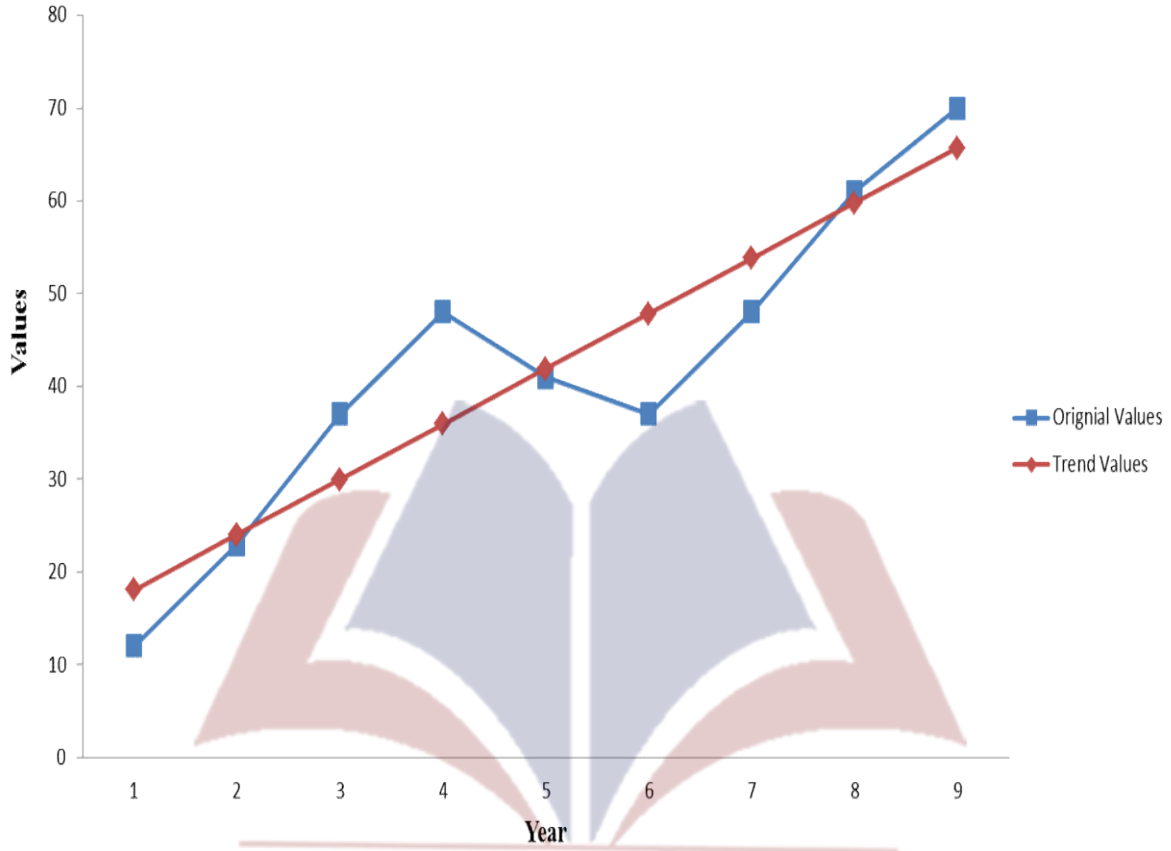
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Graph



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Lecture 9

- Analysis of Method of Least Square
- Detrending
- Analyzing the seasonal variations.

Example

Determine the trend line by the least-squares method from the following data. Plot the actual values and the linear trend on the same graph.

Two normal equations

$$a = \frac{\sum Y}{n}$$

$$b = \frac{\sum XY}{\sum X^2}$$

Year (t)	Coded year (X)	Y	XY		Trend
1945	-4	3	-12	16	2.2
1946	-3	6	-18	9	3.9
1947	-2	2	-4	4	5.6
1948	-1	10	-10	1	7.3

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1949	0	7	0	0	9.0
1950	1	9	9	1	10.7
1951	2	14	28	4	12.4
1952	3	12	36	9	14.1
1953	4	18	72	16	15.8
Total	0	81	101	60	81.0

$$a = \frac{\sum Y}{n} = \frac{81}{9} = 9$$

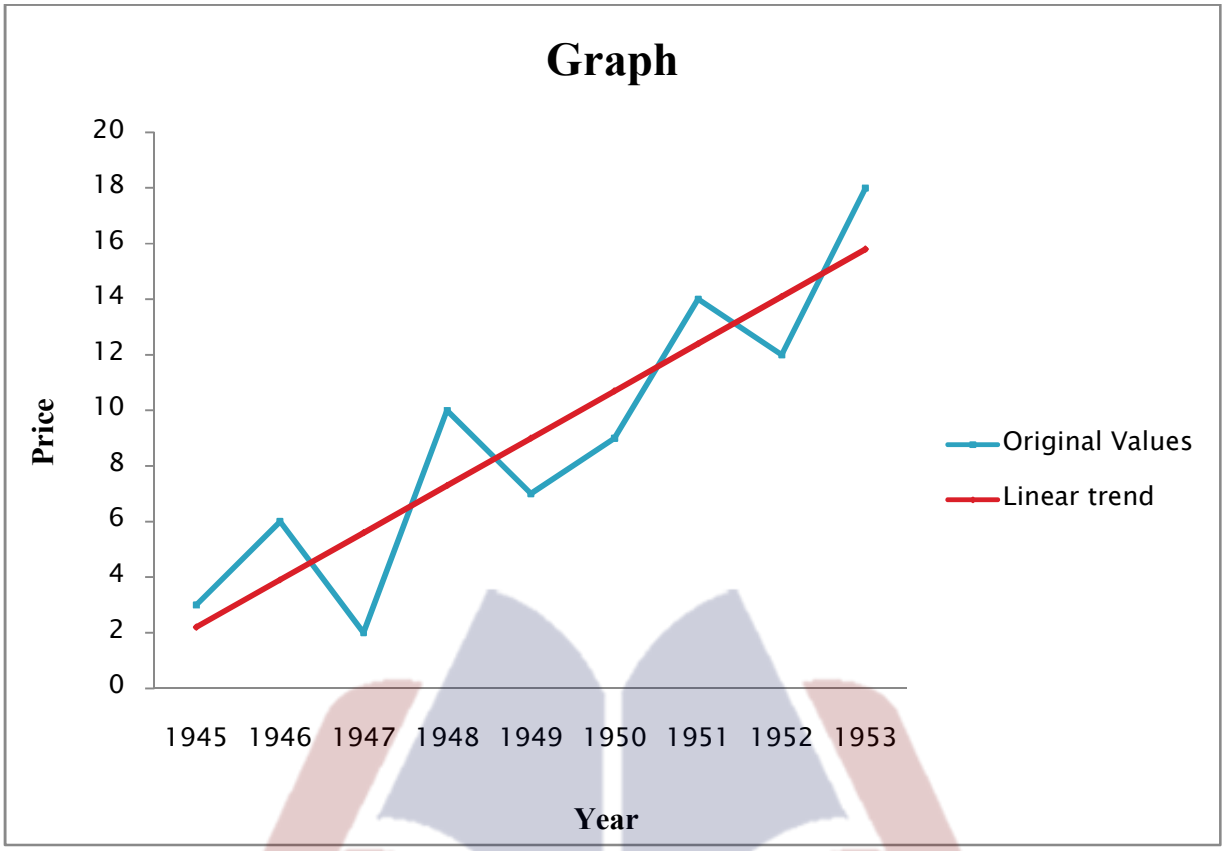
$$b = \frac{\sum XY}{\sum X^2} = \frac{101}{60} = 1.7.$$

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$$\hat{Y}_t = 9 + 1.7X$$

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Example

The following time series are the annual profits in thousands of rupees in a certain business.

Year	1977	1978	1979	1980	1981	1982	1983
Profit	88	101	105	91	113	120	132

- Use the method of least-squares to fit a straight line trend and make an estimate of the profits in 1985.
- Fit a parabola trend.
- Determine which is the better fitting trend.

The equation of the straight line:

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$$Y_t = a + bX$$

Two normal equations

$$\sum Y = na$$

$$\sum XY = b\sum X^2$$

2. The Equation of the quadratic parabola line:

$$Y_t = a + bX + cX^2$$

Three normal equations

$$\sum Y = na + c\sum X^2$$

$$\sum XY = b\sum X^2$$

$$\sum X^2Y = a\sum X^2 + c\sum X^4$$

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Year	Y	X	X^2	X^4	XY	X^2Y	Y^2
1977	88	-3	9	81	-264	792	7744
1978	101	-2	4	16	-202	404	10201
1979	105	-1	1	1	-105	105	11025

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1980	91	0	0	0	0	0	8281
1981	113	1	1	1	113	113	12769
1982	120	2	4	16	240	480	14400
1983	132	3	9	81	396	1188	17424
Total	750	0	28	196	178	3082	81844

1. The Equation of the straight line:

$$a = \frac{\sum Y}{n} = \frac{750}{7} = 107.14$$

$$b = \frac{\sum XY}{\sum X^2} = \frac{178}{28} = 6.38$$

$$\hat{Y}_t = 107.14 + 6.38X$$

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2. The Equation of the quadratic parabola line:

$$b = \frac{\sum XY}{\sum X^2} = \frac{178}{28} = 6.36$$

$$7a + 28c = 750$$

$$28a + 196c = 3082$$

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$$7a + 28c = 750 \qquad (1)$$

$$28a + 196c = 3082 \qquad (2)$$

Multiply equ (1) by 4, and subtract from equ (2), we get

$$\begin{array}{r} 28a + 112c = 3000 \\ \underline{-28a + -196c = -3082} \\ -84c = -82 \end{array} \qquad c = \frac{-82}{-84} = 0.976$$

put the value of c in equation (1), *we get*

$$7a + 28(0.976) = 750$$
$$7a + 27.328 = 750$$
$$7a = 722.672$$
$$a = 103.24$$

$$\hat{Y}_t = 103.24 + 6.36X + 0.976X^2$$

3. Determine a better fitting trend

i. In case of straight line trend.

$$\sum (Y - \hat{Y}_t)^2 = \sum Y^2 - a\sum Y - b\sum XY$$

$$\sum (Y - \hat{Y}_t)^2 = 81844 - (107.14)(850) - (6.38)(178)$$
$$= 81844 - 80355 - 1135.64 = 353.36$$

ii. In case of straight line trend.

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$$\sum (Y - \hat{Y}_t)^2 = \sum Y^2 - a\sum Y - b\sum XY - c\sum X^2 Y$$

$$\begin{aligned}\sum (Y - \hat{Y}_t)^2 &= 81844 - (103.24)(750) - (6.38)(178) - (0.976)(3.082) \\ &= 81844 - 77430 - 1135.64 - 3008.03 \\ &= 270.33\end{aligned}$$

ii. The parabolic trend is the better fit as it has the smaller sum of squares of residuals.

Example

The production of vegetables ghee ('000s tons) in Pakistan is given below:

Year	Production
1970-71	136
1971-72	162
1972-73	187
1973-74	225
1974-75	272
1975-76	277
1976-77	322

Fit a second degree parabola to the data and estimate the production for 1978-79.

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The Equation of the quadratic parabola line:

$$Y_t = a + bX + cX^2$$

Three normal equations

$$\sum Y = na + c\sum X^2$$

$$\sum XY = b\sum X^2$$

$$\sum X^2Y = a\sum X^2 + c\sum X^4$$

Year	Production Y	X					
1970-71	136	0	0	0	0	0	0
1971-72	162	1	1	1	1	162	162
1972-73	187	2	4	8	16	374	748
1973-74	225	3	9	27	81	675	2025
1974-75	272	4	16	64	256	1088	4352
1975-76	277	5	25	125	625	1385	6925
1976-77	322	6	36	216	1296	1932	11592

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Total	1581	21	91	441	2275	5616	25804

$$7a + 21b + 91c = 1581$$

$$21a + 91b + 441c = 5616$$

$$91a + 441b + 2275c = 25804$$

$$7a + 21b + 91c = 1581 \qquad (1)$$

$$21a + 91b + 441c = 5616 \qquad (2)$$

$$91a + 441b + 2275c = 25804 \qquad (3)$$

Multiply equ (1) by 3, and subtract from equ (2), we get

$$\begin{array}{r} 21a + 63b + 273c = 4743 \\ \underline{-21a + -91b + -441c = -5616} \\ -28b - 168c = -873 \end{array}$$

$$28b + 168c = 873 \qquad (4)$$

Multiply equ (1) by 13, and subtract from equ (3), we get

$$\begin{array}{r} 91a + 273b + 1183c = 20553 \\ \underline{-91a + -441b + -2275c = -25804} \\ -168b - 1092c = -5251 \end{array}$$

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$$168b + 1092c = 5251 \qquad (5)$$

Multiply equ (4) by 6, and subtract from equ (5), we get

$$\begin{array}{r} 1008b + 1008c = 5238 \\ \underline{-168b + -1092c = -5251} \\ -84c = -13 \end{array} \qquad c = \frac{-13}{-84} = 0.155$$

put the value of c in equation (4), *we get*

$$28b + 168(0.155) = 873$$
$$28b + 26.04 = 873$$
$$28b = 846.96$$
$$b = 30.25$$

Put the value of b and c in equation (1), we get

$$7a + 21b + 91c = 1581$$
$$7a + 21(30.25) + 91(0.155) = 1581$$
$$7a + 635.25 + 14.105 = 1581$$
$$7a + 649.355 = 1581$$
$$7a = 931.645$$
$$a = 133.09$$

$$\hat{Y}_t = 133.09 + 30.25X + 0.155X^2$$

Estimated production for 1978-79 is obtained by putting the value of X=8 in the fitted equation. Therefore

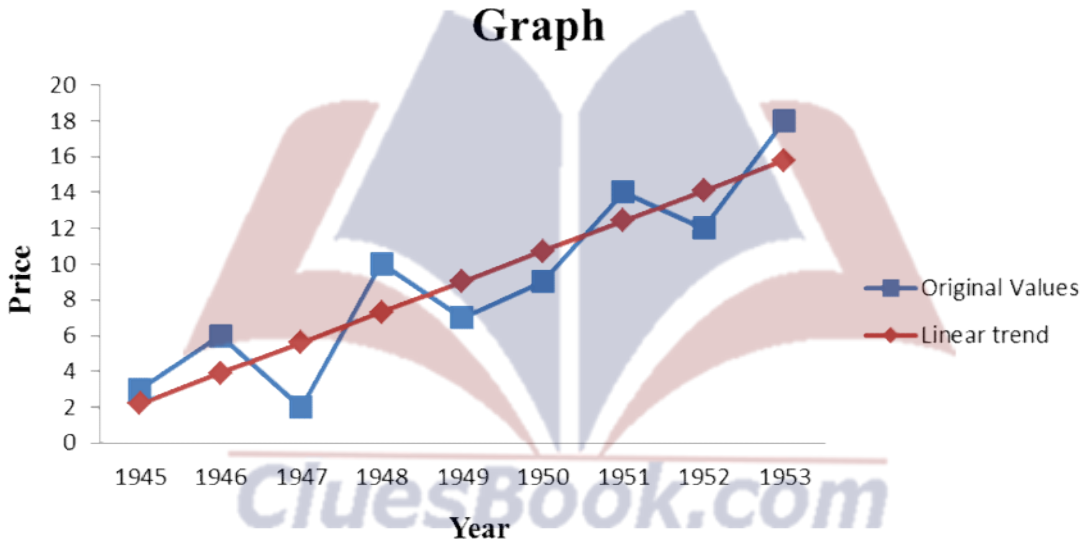
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$$\widehat{Y}_t = 133.09 + 30.25(8) + 0.155(8)^2 = 385$$

Detrending

- The process of removing the trend component from the observed time series is called "Detrending".
- A detrended time series is also known as stationary time series.
- The data in a detrended series would fluctuate around the horizontal line as displayed by the line in the following figure:



- The trend component is removed by computing either the deviations from the trend or the ratios to trend depending on whether the components follow the additive or the multiplicative model.
- When the relationship is additive, we subtract the trend values from the original observations.
- If the model is multiplicative, we divide each of the original observations by the trend values. The ratios so obtained are expressed as percentages.

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Analysing the seasonal variations

- Having removed the trend component from the time series, we are left with the deviations or ratios which are averaged for each season or month to measure the seasonal variation.
- The deviations are called seasonal differences while the ratios expressed as percentages may be called the seasonal relatives.

$$Seasonal\ relatives = \frac{original\ Y\ observations}{corresponding\ trend\ values} \times 100.$$

- A measure of seasonal variation is usually computed in index form, is called a seasonal index.
- To compute the seasonal indices, the components of the time series are assumed to follow the multiplicative model.
- Simple average of monthly or quarterly values over a period of years are known as the seasonal averages.
- When the seasonal averages are expressed as percentages of the average of all the seasonal averages, are called seasonal indices.

$$Seasonal\ Index = \frac{Seasonal\ Average}{Grand\ Average} \times 100.$$

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- A seasonal index may also be computed from weekly or daily data.

Methods for computing the Seasonal index

- The Percentage-of-Annual Average Method.
- The Ratio-to-Moving-Average Method.
- The Ratio-to-Trend Method.
- The Link-Relative Method.

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Percentage-of-Annual Average Method

- To Eliminate the effect of the trend, compute the simple averages for each year and divide each of the given monthly or quarterly observations by the corresponding annual-averages.
- To take the average of percentages with a view to removing the cyclical and irregular variations and computing the seasonal indices. For this purpose, sort out these percentages by months or quarters and find the monthly or quarterly average percentages using either the mean or the median.
- In case of mean, discard the extreme percentages under each month or quarter. If the 12 monthly or the 4 quarterly average percentage d not average to 100, adjust them by multiplying each of them by a suitable factor that will make the average of all the percentages equal 100. The resulting 12 or 4 percentages are the required seasonal indices.

Example

- Compute the seasonal indices for the following data, using the percentage-of-annual-average method.

Year	Quarters			
	I	II	III	IV
1979	72	98	79	106
1980	79	122	101	143
1981	94	141	128	160

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1982	125	143	135	187

Annual Averages

Year	**Quarters**				Annual or yearly	
	I	II	III	IV	Total	Average
1979	72	98	79	106	355	88.75
1980	79	122	101	143	445	111.25
1981	94	141	128	160	523	130.75
1982	125	143	135	187	590	147.5

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Seasonal Indices

Year	**Quarters**				Total
	I	II	III	IV	
1979	81.13	110.42	89.01	119.44	
1980	71.01	109.66	90.79	128.54	
1981	71.89	107.84	97.90	122.37	

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1982	84.75	96.65	91.53	126.78	
Total	308.78	424.87	369.23	497.13	
Mean	77.20	106.22	92.31	124.28	400.01

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Lecture 10

- Analysis of Ratio-to-Moving-Average Method
- Analysis of Ratio-to-Trend Method
- Analysis of Link-Relative Method

Ratio-to-Moving Average Method

- This method is most frequently used for the computation of seasonal index numbers.
- The first step is to eliminate the trend component. To this end, divide the original observations for each month or quarter by the corresponding 12-month or 4-quarter centered moving average and express the result as a percentage, i.e. compute the seasonal relative for each month or quarter.
- It is worthwhile to note that each monthly or quarterly value is assumed to consist of the product of the effects of T, C, S and I components, and each moving average is a measure of the combined effect of trend and cyclical components, i.e. $T \times C.$
- Dividing the original data by the corresponding moving averages and then multiplying by 100, an estimate of the effects of seasonal and irregular components combined is obtained; i.e.

$$\frac{original\ data}{moving\ average} \times 100 = \frac{TCSI}{TC} \times 100 = SI\ (\text{Seasonal relatives})$$

- The next step is to remove the effects of the irregular component in order to obtain seasonal indices. To achieve this end, arrange the seasonal relatives by months or quarters and find the monthly or quarterly averages, using either the mean or the median.

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- If mean is to be used, compute a modified mean by discarding the unusually large or small seasonal relative under each month or quarter so that the average is not distorted.
- If these monthly or quarterly averages do not average to 100, then adjust them by multiplying each median or modifies mean estimate of seasonal index by the correction factor that will make the average of all the indices equal 100.

Example

Compute the seasonal indices for the four quarters by the method of ratio-to-moving averages.

Year	**Quarters**			
	I	II	III	IV
1949	105	77	68	95
1950	107	83	74	106
1951	117	99	86	112

Year	Y-Values	4-quarter moving totals	4-quarter centered moving total	4-quarter centered moving averages	Seasonal relatives
1949-I	105	345	692	86.5	78.6
		347	700	87.5	108.6
II	77	353	712	89.0	120.2
		359	729	91.1	91.1
III	68				

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IV	95	370	750	93.8	78.9
1950-I	107	380	776	97.0	109.3
II	83	396	804	100.5	116.4
III	74	408	822	102.8	96.3
IV	106	414			
1951-I	117				
II	98				
III	86				
IV	112				

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Seasonal Indices

Year	Quarters				Total
	I	II	III	IV	
1949			78.6	108.6	
1950	120.2	91.1	78.9	109.3	
1951	116.4	96.3			

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Total	236.6	187.4	157.5	217.9	
Mean S.I	118.30	93.70	78.75	108.95	399.7

Ratio-to-Trend Method

- In this method, the trend values are obtained for each time period by fitting a least-squares trend line either to the observed time series data or to the annual averages.
- The rest of the computational procedure is the same as that of the ratio-to-moving-average method as the seasonal index computed by it includes cyclical and irregular variations.

Example

Compute seasonal indices for the four quarters by the ratio-to-trend method for the following data.

Year	**Quarters**			
	I	II	III	IV
1961	122	125	118	117
1962	119	114	114	109
1963	105	99	93	89
1964	86	80	83	84

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Year	Y-Values	X		XY	Trend
1961-I	122	-15	225	-1830	127.56
II	125	-13	169	-1625	124.36
III	118	-11	121	-1298	121.16
IV	117	-9	81	-1053	117.96
1962-I	119	-7	49	-833	114.76
II	114	-5	25	-570	111.56
III	114	-3	9	-342	108.36
IV	109	-1	1	-109	105.16
1963-I	105	1	1	105	101.96
II	99	3	9	297	98.76
III	93	5	25	465	95.56
IV	89	7	49	623	92.36
1964-I	86	9	81	774	89.16
II	80	11	121	880	85.96
III	83	13	169	1079	82.76

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IV	84	15	225	1260	79.56

The Equation of the straight line:

$$Y_t = a + bX$$

Two Normal Equations:

$$a = \frac{\sum Y}{n} = \frac{1657}{16} = 103.56$$

$$b = \frac{\sum XY}{\sum X^2} = \frac{-2177}{1360} = -1.60$$

$$\hat{Y}_t = 103.56 - 1.60X$$

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Seasonal Indices

Year	Quarters				Total
	I	II	III	IV	
1961	95.64	100.51	97.39	99.19	
1962	103.69	102.19	105.20	103.65	
1963	102.98	100.24	97.32	96.36	

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1964	96.46	93.07	100.29	105.58	
Total	398.77	302.94	295.00	404.78	
Mean	99.69	100.98	98.33	101.20	400.20
S.I	99.64	100.93	98.28	101.15	400

Link-Relative Method

- Compute the link-relatives by expressing each monthly or quarterly value as a percentage of the preceding monthly or quarterly value.
- Arrange the link-relatives by months or quarters and find an appropriate average of these relatives for each month or quarter.
- Convert the average (median or mean) relatives into a series of chain relatives by setting the value of January or the first quarter as 100, and carrying the process to include the first unit of the next period.
- A discrepancy due to trend increment positive an negative) exists between the chain relative for the first January or quarter and that for the next period. Adjust the chain relatives for the trend component by subtracting one-twelfth of the discrepancy, from the value of February, two-twelfth from the value of March and so on or by one-fourth of the discrepancy from the relative of second quarter, two-fourth from the third quarter relative and three fourth from the fourth quarter relative.
- To obtain the seasonal indices, reduce the adjusted chain relatives to the same level as January or the first quarter by multiplying each of the adjusted chain relatives by the correction factor that will make the average of the indices equal 100.

Example

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Obtain the seasonal indices for the following data by using the link-relative method.

Year	**Quarters**			
	I	II	III	IV
1948	71	89	106	78
1949	71	90	108	79
1950	73	91	111	81
1951	76	97	122	89

Link-Relative

Year	**Quarters**			
	I	II	III	IV
1948		125.4	119.1	73.6
1949	91.0	126.8	120.0	73.1
1950	92.4	124.7	122.0	73.0
1951	93.8	127.6	125.8	73.0
Median	92.4	126.1	121.0	73.1

Chain relatives

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Quarter	I	II	III	IV	I
Chain Relative	100	126.1	152.6	111.6	103.1

The chain relative for the first quarter works out to be 103.1 which as a matter of fact, ought to have been 100. This increase of 3.1 is due to the trend component present in the data. An adjustment for the trend therefore become necessary. Since the difference is positive so we subtract one-fourth of this form the second quarter, two-fourth from the third quarter and three-fourth from the fourth quarter.

Seasonal Indices

Quarter	I	II	III	IV	Total
Adjusted Chain Relative	100	125.32	151.05	109.28	485.65
Seasonal Index	82.4	103.2	124.4	90.0	400.0

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Lecture 11

- Deseasonalization of Data
- Examples

Deseasonalization of Data

- The process of removing the seasonal component from a time series is called deseasonalization or seasonal adjustment of data and the time series thus obtained is called the deseasonalization or "seasonally adjusted time series.
- To get deseasonalized data, we divide (considering multiplicative model) each value in the original data by the corresponding value of seasonal index and multiply the result by 100. Thus

$$Deseasonalized\ data = \frac{period's\ original\ value}{period's\ seasonal\ index} \times 100$$

$$= \frac{TCSI}{S} \times 100 = TCI \times 100$$

- Thus the deseasonalized data contain the contain the effect of trend, cyclical and irregular components.

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Example

Obtain the seasonal indices by the ratio-to-trend method and by fitting the least-squares trend line to the annual averages of the following data, showing the amount of the money (*£ 000,000)* spent upon passenger travel in the United Kingdom at levels of fares, and charges current during the periods given. Also deseasonalized the data for 1950 year.

Year	**Quarters**			
	I	II	III	IV
1948	71	89	106	78

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1949	71	90	108	79
1950	73	91	111	81
1951	76	97	122	89

Fit a straight line by the method of least-squares to the annual average:

Year	X	Y	XY	X^2
1948	-3	86	-258	9
1949	-1	87	-87	1
1950	1	89	89	1
1951	3	96	288	9
Total			32	20

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$$Y_t = a + bX$$

Two normal equations:

$$a = \frac{\sum Y}{n} = \frac{358}{4} = 89.5$$

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$$b = \frac{\sum XY}{\sum X^2} = \frac{32}{20} = 1.6$$

$$\hat{Y}_t = 89.5 + 1.6X$$

- The Quarterly trend Values:

$$\hat{Y} = 89.5 + 1.6X$$

Year	**Quarters**			
	I	II	III	IV
1948	83.5	84.3	85.1	85.9
1949	86.7	87.5	88.3	89.1
1950	89.9	90.7	91.5	92.3
1951	93.1	93.9	94.7	95.5

- Deseasonalized data for 1950 year are shown below:

Quarter	Amount of Money	Seasonal Index	Deseasonalized Data
I	73	81.2	90

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II	91	100.3	91
III	111	128.8	86
IV	81	93.2	87

Example

Compute the seasonal indices by using the ratio-to-moving-average method for the following data. Also deseasonalized the data for 1980 year.

Year	**Quarters**			
	I	II	III	IV
1979	72	98	79	106
1980	79	122	101	143
1981	94	141	128	160
1982	125	143	135	187

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Year	Y-Values	4-quarter centered moving averages	Seasonal relatives
1979-I	72	89.6	88.2
II	98	93.5	113.4

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III	79	99.2	79.6
IV	106	106.6	114.4
1980-I	79	113.1	89.3
II	122	117.4	121.8
III	101	123.1	76.4
IV	143	128.6	109.6
1981-I	94	134.6	95.1
II	141	138.8	115.3
III	128	139.9	89.3
IV	160	144.1	99.2
1982-I	125		
II	143		
III	135		
IV	187		

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Seasonal Indices

Year	Quarters	Total

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	I	II	III	IV	
1979			88.2	113.4	
1980	79.6	114.4	89.3	121.8	
1981	76.4	109.6	95.1	115.3	
1982	89.3	99.2			
Total	156.0	224.0	272.6	350.5	
Mean	78.0	112.0	90.87	116.83	397.70
S.I	78.45	112.65	91.49	117.51	400.01

- Deseasonalized data for 1980 are shown below:

Quarter	Number of bags of fertilizer (00)	Seasonal Index	Deseasonalized Data
I	79	78.45	101
II	122	112.65	108
III	101	91.40	111
IV	143	117.51	122

Example

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Fit a quadratic parabola to the following series of observations, taking the year as the independent variable.

Year	1924	1927	1930	1933	1936	1939	1942
Index of coal price	187	142	133	129	136	169	279

Use your results to estimate the value of the index for 1935.

The Equation of the quadratic parabola:

$$\hat{Y}_t = a + bX + cX^2$$

The Normal Equations for determining a, b and c are:

$$\sum Y = na + b\sum X + c\sum X^2$$

$$\sum XY = a\sum X + b\sum X^2 + c\sum X^3$$

$$\sum X^2Y = a\sum X^2 + b\sum X^3 + c\sum X^4$$

Fit a quadratic parabola line by the method of least-squares to the annual average:

Year	Y	X	X^2	X^3	X^4	XY	X^2Y
1924	187	0	0	0	0	0	0
1927	142	3	9	27	81	426	1278

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1930	133	6	36	216	1296	798	4788
1933	129	9	81	729	6561	1161	10449
1936	136	12	144	1728	20736	1632	19584
1939	169	15	225	3375	50625	2535	38025
1942	279	18	324	5832	104976	5022	90396
Total	1175	63	819	11907	184275	11574	164520

Substituting these sums, we get

$7a + 63b + 819c = 1175$ (1)

$63a + 819b + 11907c = 11574$ (2)

$819a + 11907b + 184275c = 164520$ (3)

Multiply equ (1) by 9, and subtract from equ (2), we get

$$\begin{array}{r} 63a + 567b + 7371c = 10575 \\ \underline{-63a - 819b - 11907c = -11574} \\ -252b - 4536c = -999 \end{array}$$

$$252b + 4536c = 999 \qquad (4)$$

Multiply equ (2) by 13, and subtract from equ (3), we get

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$$819a + 10647b + 154791c = 150462$$
$$\underline{-819a + -11907b + -184275c = -164520}$$
$$-1260b - 29484c = -14058$$

$$1260b + 29484c = 14058 \qquad (5)$$

Multiply equ (4) by 5, and subtract from equ (5), we get

$$1260b + 22680c = 4995$$
$$\underline{-1260b + -29484c = -14058}$$
$$-6804c = -9063$$

$$c = \frac{-9063}{-6804} = 1.332$$

put the value of c in equation (4), *we get*

$$252b + 4536(1.332) = 999$$
$$252b + 6041.952 = 999$$
$$252b = -5042.952$$
$$b = -20.012$$

Put the value of b and c in equation (1), we get

$$7a + 63b + 819c = 1175$$
$$7a + 63(-20.012) + 819(1.332) = 1175$$
$$7\,a - 1260.756 + 1090.908 = 1175$$
$$7a - 169.848 = 1175$$
$$7a = 1344.848$$
$$a = 192.12$$

$$\hat{Y}_t = 192.12 - 20.012X + 1.332X^2$$

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- Hence the required equation of the quadratic parabola is:

$$\hat{Y}_t = 192.12 - 20.012X + 1.332X^2$$

With the origin at 1924.

To estimate the value of the index for 1935, we put X = 11 in the required equation of the quadratic parabola and get

$$\begin{aligned}\hat{Y}_t &= 192.12 - 20.012(11) + 1.332(11)^2 \\ &= 192.12 - 220.132 + 161.172 \\ &= 133.16\end{aligned}$$

e

Example

A merchant's sales ('00s tons) of ordinary coal over a period were as shown below:

Year	Quarters			
	I	II	III	IV
1976	118	87	47	83
1977	94	73	41	68
1978	73	61	36	56

Construct seasonal indices, using the percentage of annual-average method.

Annual Averages

Year	Quarters	Annual or

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					Yearly	
	I	II	III	IV	Total	Average
1976	118	87	47	83	335	83.75
1977	94	73	41	68	276	69.00
1978	73	61	36	56	226	56.50

Seasonal Indices

Year	Quarters				Total
	I	II	III	IV	
1976	140.90	103.88	56.12	99.10	
1977	136.23	105.80	59.42	98.55	
1978	129.20	107.96	63.72	99.12	
Total	406.33	317.96	179.26	296.77	
Mean	135.44	105.88	59.75	98.92	399.99

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Lecture 12

- Introduction
- Vital Registration System
- Use of Vital Statistics
- Shortcomings of Vital Statistics
- Rates and Ratios
- Vital Ratios
- Classification of Vital Rates

Introduction to Vital Statistics

- A study of the behavior of the populations involves a study of two main aspects of a population, namely, its composition and the changes which take place during some period.
- The composition of a population is described by the distribution of people among more or less some standard categories, e.g. gender, age, marital status, etc.
- Changes in the population are the result of events like deaths and births, which add or take away some members of the population.
- Since deaths and births are very important events in human life, they are called vital events.
- Marriages, divorces, sickness, migration, etc. are also called vital events.
- The collection, presentation and analysis of vital events is called Vital Statistics.

Sources of Data

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- There are two traditional sources of data on population, namely, the census and the vital registration system.
- Vital statistics can be obtained from a census while the composition of the population can be determined from a registration system.

Population Census

- Census of population is the total process of collecting and publishing demographic, economic and social data pertaining to a specified time or times of all persons in a country of delimited territory.
- Only five census of population have been conducted in Pakistan. The first Population census was conducted in 1951 year, the second in 1961, the third in 1972, the fourth in 1981 and the fifth in 1998.
- The dates and population figures for these census are given in the following table (which on the next page).

Census	Dates of enumeration	Intercensal period since the previous census	Population enumerated (000)	Population Intercensal change
1951	Feb, 9-28	10 years	33,780	19.3
1961	Jan, 12-31	9 years and 11 months	42,188	27.2
1972	Sept, 16-30	11 years and 9 months	65,310	51.3
1981	March, 1-15	8 years and 6 months	83,782	28.3
1998	March, 1-13	17 years	132,350	42.0

Vital Registration System

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- The vital records may be defined as those dealing with births, deaths, still births (A still birth is the one in which the child shows no signs of life after being completely separated from the mother), foetal deaths (A foetal death is the death before complete expulsion from its mother), adoptions, legitimation, divorces, and separation.
- The registration of births and deaths in the rural areas of Pakistan is carried out under the Basic Democracies Order, 1959 and in urban areas under the Municipal Administration Ordinance, 1960.
- The registration of births and deaths in the rural areas is the responsibility of the union councils. The head of the family is responsible for getting the event registered either at the office of the Union Council or to report the event to the chowkidar of the village who would get the same registered at the office of the Union Council.
- The registration of births and deaths in the urban areas is carried out under the Municipal Administration Ordinance, 1960. The head of the household is responsible to get the event registered at the registration office.
- The following details are noted in the birth and death registers in respect of each event.

Birth Register	Death Register
Annual serial number	Annual serial number
Date of report of birth	Date of report of death
Date of birth	Date of death
Gender of the child	Gender of the deceased
Name of the father and grandfather of the child	Name of the father (or husband of the deceased)
Place of birth	Place of death

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Occupation, cast, religion and residence of the father	Age of the deceased
Name of the child	Name of the deceased
Name of mother	Cast, religion and residence of deceased

Use of Vital Statistics

- Records of births, deaths, marriages, divorces, etc. are of huge use to the individuals.
- Insurance companies make use of vital statistics in determining the rates of premium for different ages on the basis of expectation of life at different age-groups.
- Vital statistics provide a numerical assessment of the state of public health, hygienic conditions and availability of medical facilities.
- Vital statistics are used by various government agencies for a number of administrative purposes.
- Use of Vital Statistics in planning for economic development is necessary.
- Vital statistics are of vast use to the businessmen.
- Accurately registered and systematically collected data on vital events, e.g. the registration of people with Registration Organization (now called NADRA) and registration of births and deaths with Health department can be used to cross-check the accuracy of data obtained through population census.
- Vital statistics are necessary in demographic research.
- Medical and pharmaceutical research is carried out on the basis of mortality and natality data. Statistical methods enable planning of experiments and evaluation of results of different treatments and medicines used for curing the disease.

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Shortcomings of Vital Statistics

- There exists a large amount of under-registration of the vital events. Many deaths, marriages, divorces also remain unregistered.
- Another defect is the delayed reporting of births and deaths by the reporting agencies.
- Statistics regarding the caused of death are not reliable. Data on ages at death suffer from similar defects.
- In case of births, the age of mother and the order of birth/live-birth are not registered which is a handicap for detailed analysis of statistics.
- The basic limitation of the registration system is an almost complete lack of control over the local registrars who are neither properly trained for carrying out this duty nor their work is properly supervised.
- Data on vital statistics are sometimes misreported intentionally.

Rates and Ratios

- Vital data comprising number of births, deaths, sickness, etc. represent absolute numbers. The absolute numbers, in themselves, cannot make useful comparisons. For example, the fact that 2,000 deaths were recorded in a particular city is of no significance unless the population of city is known.
- To draw meaningful conclusions, therefore, we need relative numbers. The relative numbers are of two types-rates and ratios.
- A ratio is a single number indicating the relative size of two numbers. If “a” and “c” are two numbers, then ration between them is defined as “a/c”.
- In statistics, a ratio expresses the relation of occurrence of a given kind of event to be occurrence other event or of one kind of data to the other. Symbolically, we may define a ratio as

$$Ratio = \frac{a}{c}$$

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- Where a = number of times a given kind of event occurs.

  b = number of times another kind of event occurs.

- A ratio is usually multiplied by 100 or 1000 for ease in understanding and comparison. Some of the important ratios used in vital statistics are sex ratios and child-women ratio.

- A ratio of two numbers such that the numerator is a part of the denominator and the denominator represents the total number is called a proportion.

- A ratio of the type a/(a+b) shows proportion of "a" in "a+b". A proportion can never be greater than 1.

- For example, the proportion of bachelors upto the age of 25 years to the total population is the probability that so many persons out of the total population will be bachelors till they attain the age of 25 years.

- In vital statistics, a ratio which expresses the numerical proportion of the number of occurrence of a vital event to the total population is called a rate. Symbolically, a rate may be defined as

$$Rate = \frac{a}{a + b}$$

Where a = number of times a given event occurs

b = number of times the event does not occur

- Vital rates are usually expressed per 1000 population for ease in understanding. Vital rates are death rates, birth rates, fertility rates, reproduction rates, morbidity rates and marriage rates.

Vital Ratios

There are several ratios which are used in vital statistics depending upon the need of the study. The commonly used vital ratios are:

1. Sex Ratios

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2. Child Women Ratio

Sex Ratios

- It is defined as the ratio of the number of males to the number of females in a given population.

$$Sex\ Ratio = \frac{M}{F} \times 100$$

Where M = number of males in the population

F = number of females in the population

- For example, if the population of a city consists 2,10,326 males and 2,08,859 females, then the sex ratio is (210326/208859) x 100= 100.7. this shows that in this population the males are 1.007 times or 100.7 percent of males.
- A sex ratio of more than 100 shows that the number of males is more than the number of females. Conversely, a sex ratio of less than 100 shows that the number of females is more than the number of males.
- Sometimes we are interested in the sex ratio of a portion of the population. For example, the sex ratio at birth describes the sex composition of the live births at a specified time. It is given by:

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$$Sex\ Ratio\ at\ Birth = \frac{B_m}{B_f} \times 100$$

Where B_m = number of males live-births

B_f = number of female live-births

- For example, if in an area during a specified period or year the number of male babies born is 154 thousand and that of female babies born is 140 thousand then

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$$Sex\ Ratio\ at\ Birth = \frac{154,000}{140,000} \times 100 = 110$$

- Which shows that there are 110 male babies for 100 female babies at the time of birth in the area during the period or year.
- The sex ratios at different age-groups may also be computed.

Age	Males	Females
0 – 9	64,56,106	61,73,394
10 – 19	43,14,011	33,27,908
20 – 39	56,77,437	50,64,890
40 – 59	31,57,925	25,36,835
60 & over	15,62,568	11,71,365
All ages	2,11,68,047	1,82,74,392

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Age	Males	Females	Sex Ratio
0 – 9	64,56,106	61,73,394	109.4
10 – 19	43,14,011	33,27,908	120.6
20 – 39	56,77,437	50,64,890	112.1

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40 – 59	31,57,925	25,36,835	124.5
60 & over	15,62,568	11,71,365	133.4
All ages	2,11,68,047	1,82,74,392	115.8

Example

Calculate sex ratios by age for 1961 Population Census of Pakistan and examine your results critically.

Age-Group	Population (000)	
	Males	Females
All ages	21168	18274
0 – 4	3301	3157
5 – 9	3456	3016
10 – 14	2098	1711
15 – 19	1916	1617
20 – 24	1635	1448
25 – 29	1586	1456
30 – 34	1330	1202
35 – 39	1126	958

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40 – 44	1040	871

Age-Group	Population (000)		Sex Ratios
	Males	Females	
All ages	21168	18274	115.8367
0 – 4	3301	3157	104.5613
5 – 9	3456	3016	114.5889
10 – 14	2098	1711	122.6184
15 – 19	1916	1617	118.491
20 – 24	1635	1448	112.9144
25 – 29	1586	1456	108.9286
30 – 34	1330	1202	110.6489
35 – 39	1126	958	117.5365
40 – 44	1040	871	119.403

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Child-Women Ratio

- Child-women ratio is the ratio of children to the women in census population of a country or a geographic region.
- It gives a measure of incidence of child-bearing in the population of adult women.

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- It is the ratio of children under 5 years of age to the women of child-bearing age in the population of adult women, i.e. between the ages 15 – 44 years or sometimes taken as 15 – 49 years. Thus

$$Child-Women\ Ratio = \frac{P_{0-4}}{f_{15-44}} \times 100$$

Where P_{0-4} = number of children of both sexes under 5 years of age

f_{15-44} = number of females between ages 15 – 44 (sometimes taken as 15-49)

- According to the 1961 Census of Population, the number of children below 5 years of age and the female population between the ages 15 – 44 in Pakistan were enumerated to be 6,457 thousand and 6,627 thousand respectively. Thus the child-women ratio is

$$Child-Women\ Ratio = \frac{6,457000}{6,627000} \times 100 = 97.43$$

Classification of Vital Rates

- The rates used in vital statistics may be classified as observed and theoretical rates.
- The observed rates are simple in nature and are computed directly from observed or actual data.
- Theoretical rates are complex in nature and are calculated from some theoretical population known as standard population.
- The observed rates particularly the death and birth rates may be classified as crude and specific rates.
- A crude rate expresses the relation of the vital events occurring during a year to the mid-year population during the same year.

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- A specific rate expresses the relation of vital events in some specific class to the population of that class.
- Mortality Rates (Death Rates)

(a) Observed (i) Crude death rate (ii) Specific death rate

(b) Theoretical: Standardized death rate or adjusted death rate

- Natality Rates (Birth Rates)

(a) Observed (i) Crude birth rate (ii) Specific birth rate

(b) Theoretical: Standardized birth rate or adjusted birth rate

- Fertility Rates (Observed)

(i) General fertility rate (ii) Specific fertility rate

- Reproduction Rate (Observed)

(i) Gross reproduction rate (ii) Net reproduction rate

- Morbidity or sickness rates
- Marriage rates
- Divorce rates

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Crude Death Rate

- The simplest and the most commonly used measure of mortality is the crude death rate (C.D.R).
- It is defined as the number of death per 1000 mid-year population of an area during a year.
- Crude death rate is the ratio of registered deaths in an area during a year to the mid-year estimated population of the same area during the same year usually expressed per 1000. Symbolically

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$$C.D.R = \frac{D}{P} \times 1000$$

Where D = number of registered deaths in an area during a year.

P = mid-year estimated population of the are for the same year

Example

Suppose the number of registered deaths for a country during the calendar year 1960 was 10,063 and the estimated population of the country on 1st July, 1960 was 1,171, thousand. Then the crude death rate of the country for the year 1960 is

$$C.D.R = \frac{10,063}{1,171,000} \times 1000 = 8.6 \; per \; thousand$$

Specific Death Rate

- The crude death rate gives only a general indication of the level of mortality and its changes.
- The mortality rate of a population is greatly affected by a number of factors, particularly the age and sex distribution of the population. For example, the death rate is high at early and old ages but low at middle ages.
- A crude death rate fails to make comparisons between different ages, areas, etc.
- Specific death rate may be defined as the number of deaths per 1000 population of a specific group or class; the specification being made with respect to age, sex, marital status, occupation, religion, etc. The specific death rates are the age-specific and sex-specific death rates.

The age-specific death rate, denoted by M_i, is computed as

$$M_i = \frac{d_i}{p_i} \times 1000$$

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Where d_i = number of deaths during the year in the ith age-group

p_i = mid-year population in the ith age-group

The death rates computed separately for males and females will be sex-specific death rates. They can be computed by a formula similar to the one given above. When the age-specific death rates are computed separately for males and females, they are called age-sex specific death rates.

Example

Find crude death rate and age-specific death rates for the following data.

Age (year)	Population	Number of Deaths
Below 1	51175	4456
1 – 4	197772	1476
5 – 14	387718	438
15 – 24	233350	303
25 – 34	161926	336
35 – 44	121119	414
45 – 54	86679	540
55 – 64	52185	758
65 – 74	27045	1001
75 & over	15205	1544

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Age-specific Death Rate

Age (year)	Population	Number of Deaths
Below 1	51175	4456
1 – 4	197772	1476
5 – 14	387718	438
15 – 24	233350	303
25 – 34	161926	336
35 – 44	121119	414
45 – 54	86679	540
55 – 64	52185	758
65 – 74	27045	1001
75 & over	15205	1544

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Crude Death Rate

Age (year)	Population	Number of Deaths
Below 1	51175	4456
1 – 4	197772	1476

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5 – 14	387718	438
15 – 24	233350	303
25 – 34	161926	336
35 – 44	121119	414
45 – 54	86679	540
55 – 64	52185	758
65 – 74	27045	1001
75 & over	15205	1544
Total	1334174	11266

$$C.D.R = \frac{D}{P} \times 1000$$

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$$C.D.R = \frac{11266}{1334174} \times 1000 = 8.44 \; per \; thousand$$

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Lecture 13

- Observed Mortality Rate
- Theoretical Mortality Rate

Example

Calculate age-sex specific death rates for the following data for selected age-groups.

Age-group (years)	**Midyear population**		**No. of Deaths**	
	Males	Females	Males	Females
10 – 14	90,229	69,379	1,043	959
15 – 19	83,442	56,371	931	917
20 – 29	164,377	103,014	1,111	914
30 – 39	111,878	68,724	1,039	885

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40 – 49	69,008	48,777	1,017	812
50 – 59	39,732	29,420	1,008	774

Age Specific Death Rate

Age-group (years)	Age Specific death rates per 1000 person per year	
	Males	Females
10 – 14	11.55948	13.82263
15 – 19	11.15745	16.26723
20 – 29	6.758853	8.87258
30 – 39	9.286902	12.8776
40 – 49	14.73742	16.64719
50 – 59	25.36998	26.30863

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Infant Mortality Rate

- It is defined as the ratio of registered death of infants (under one year of age) in an area during a specified year to the total live-births registered in the same year.

$$I.M.R = \frac{d_0}{B} \times 1000$$

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Where I.M.R stands for infant mortality rate.

d_0 denotes the number of deaths of infants registered in an area during a specified year

B denotes the number of live-births registered during the same year in the same area.

The infant mortality rate does not provide an accurate measure of the risk of death during the first year of life because

- Infants are usually under-enumerated.
- The babies who die immediately after birth, are often not registered as live births.
- Some of the deaths under one year of age during a specified year must have been of infants who had been born in the preceding specified year.
- Sometimes infant deaths rate are not registered from stillbirths and abortions.

Standardized Death Rates

- The crude death rates of two localities or in two occupations cannot be compared because mortality rates differ with age, gender, climate, occupation, etc. To eliminate such effects, we compute the standardized death rates.
- Standardized death rates can be computed either by direct method or indirect method.

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Direct Method

- The direct method for computing the standardized death rate consists in calculating the number of deaths that would be expected to occur in a standard population, if the age-specific death rates of the given population were to apply and dividing the sum of the expected deaths by the standard population.

$$S.D.R = \frac{Expected\ deaths\ in\ s\tan dard\ \ population}{Total\ S\tan dard\ \ population} \times 1000$$

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$$S.D.R = \frac{\sum \frac{d_i}{p_i} \times P_i}{\sum P_i} \times 1000$$

Where S.D.R stands for standardized death rate.

d_i denotes the number of deaths occurring in the given population in the ith age-group during a specified year.

p_i denotes the midyear population figures in the given population in ith age-group during the same year

P_i denotes the midyear population figure in the standard population in ith age-group in the same year.

If the death rates are adjusted both for age and sex, the expected deaths are calculated as follows:

$$Expected\ deaths\ for\ males\ (m) = \frac{d_{im}}{p_{im}} \times P_{im}$$

$$Expected\ deaths\ for\ females\ (m) = \frac{d_{if}}{p_{if}} \times P_{if}$$

Indirect Method

- When the age-specific death rates for the given population are not available or there are fluctuations in age-specific death rates, we determine the number of deaths that would be expected in the given population, assuming that the age specific death rates of the standard population had prevailed for one year in the given population.

$$S.D.R = \frac{Observed\ deaths\ in the\ given\ population}{Expected\ deaths\ \text{in the}\ given\ population} \times C.D.R\ of\ s\tan dard\ population$$

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$$Expected\ Death = \frac{D_i}{P_i} \times p_i$$

D_i denotes the number of deaths occurring in the ith age-group in the standard population during a year.

p_i denotes the midyear population figures in the given population in ith age-group during of the given population during the same year.

P_i denotes the midyear population figure in the ith age-group of the given population in the same year.

The Expected death for both males and females are calculated separately in a similar way and the formula applied accordingly.

$$S.D.R = \frac{\sum d_{im} + \sum d_{if}}{\sum (\frac{D_{im}}{P_{im}} \times p_{im}) + \sum (\frac{D_{if}}{P_{if}} \times p_{if})}$$

Example

Given the following data for two districts:

Age-group (years)	District A		District B	
	Population	Death	Population	Death
0 – 14	500	2	400	1
15 – 29	2,000	8	300	1

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30 – 44	2,000	12	1,000	5
45 – 59	1,000	10	2,000	18
60 – 74	500	20	2,000	70
75 & over	100	15	400	50

Compute Crude Death Rates for Districts A and B. Considering District A as the standard population, calculate the standardized death rate for District B, using direct method.

Age-group (years)	District A			District B		
	Population	Death	A.S.D.R	Population	Death	A.S.D.R
0 – 14	500	2	4	400	1	2.5
15 – 29	2,000	8	4	300	1	3.333333
30 – 44	2,000	12	6	1,000	5	5
45 – 59	1,000	10	10	2,000	18	9
60 – 74	500	20	40	2,000	70	35
75 & over	100	15	150	400	50	125

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Total	6100	67		6100	23.8	

$$Crude\ Death\ rate\ for\ District\ A = \frac{Number\ of\ Deaths}{Number\ of\ Population} \times 1000$$

$$= \frac{67}{6100} \times 1000 = 10.98$$

$$Crude\ Death\ rate\ for\ District\ B = \frac{Number\ of\ Deaths}{Number\ of\ Population} \times 1000$$

$$= \frac{145}{6100} \times 1000 = 23.77$$

$$Expected\ Death\ Rate = \frac{d_i}{p_i} \times P_i$$

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Age-group (years)	Expected Death in Standard Population as experienced in District A
0 – 14	$\frac{1}{400} \times 500 = 1$
15 – 29	$\frac{1}{300} \times 2,000 = 7$

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30 – 44	$\frac{5}{1,000} \times 2,000 = 10$
45 – 59	$\frac{18}{2,000} \times 1,000 = 9$
60 – 74	$\frac{70}{2,000} \times 500 = 18$
75 & over	$\frac{50}{400} \times 100 = 12$

$$S.D.R = \frac{\sum \frac{d_i}{p_i} \times P_i}{\sum P_i} \times 1000$$

$$= \frac{57}{6100} \times 1000 = 9.3$$

Example

Calculate the crude and standardized death rates from the following data.

Age-group (years)	District A		Standard	
	Population (p)	Death (d)	Population (P)	Death (D)
0 – 9	20,210	728	231,428	6,960
10 – 24	29,875	213	302,170	1,972

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25 – 44	37,294	462	293,149	4,759
45 – 64	16,739	741	150,122	7,030
65 & over	4,925	823	46,500	8,314

$$Crude\ Death\ rate\ for\ District\ A = \frac{Number\ of\ Deaths}{Number\ of\ Population} \times 1000$$

$$= \frac{2,967}{109,043} \times 1000 = 27.21$$

$$Crude\ Death\ rate\ for\ S\tan dard = \frac{Number\ of\ Deaths}{Number\ of\ Population} \times 1000$$

$$= \frac{29,035}{1023,369} \times 1000 = 28.37$$

$$Expected\ Death\ Rate = \frac{d_i}{p_i} \times P_i$$

Direct Method

$$Expected\ Death\ Rate = \frac{D_i}{P_i} \times p_i$$

Indirect Method

Age-group (years)	Expected Death in Standard Population	Expected Death in District A
0 – 9	8,336	608
10 – 24	2,154	195

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25 – 44	3,632	605
45 – 64	6,646	784
65 & over	7,770	881
	28,538	3,073

Standardized Death Rate (Direct) = $$S.D.R = \frac{\sum \frac{d_i}{p_i} \times P_i}{\sum P_i} \times 1000$$

$$= \frac{28,538}{1023,369} \times 1000 = 27.89$$

Standardized Death Rate (Indirect) =

$$S.D.R = \frac{Observed\ deaths\ in the\ given\ population}{Expected\ deaths \text{ in the } given\ population} \times C.D.R\ of\ s\tan dard\ population$$

$$= \frac{2,967}{3,073} \times 28.37 = 27.39$$

Example

Calculate the standardized death rates for District A, taking the population of the Province as standard from the following data using

1) Direct Method 2) Indirect Method

Age-	Population of the	Death in the	Population of	Deaths in

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group (years)	Province		Province		District A		District A	
	Male	Female	Male	Female	Male	Female	Male	Female
0 – 9	19,440	17,640	658	592	1,405	980	52	45
10 – 29	27,360	25,800	305	245	1,140	1,075	15	12
30 – 49	15,645	13,860	245	178	745	660	13	10
50 – 59	3,825	3,150	114	87	228	184	7	5
60 & over	3,730	3,730	228	178	202	141	15	13

Age-group (years)	Expected Death in Standard Population		Expected deaths in District A on the basic of deaths in Standard Population (Province)	
	Male	Female	Male	Female
0 – 9	967	810	35	33

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10 – 29	360	283	13	10
30 – 49	273	210	12	8
50 – 59	117	86	7	5
60 & over	277	235	12	10
Total	1994	1629	79	66

Standardized Death Rate (Direct Method) =

$$S.D.R = \frac{\sum \frac{d_{im}}{p_{im}} P_{im} + \sum \frac{d_{if}}{p_{if}} P_{if}}{\sum (P_{im} + P_{if})} \times 1000$$

$$= \frac{1994 + 1629}{70000 + 6300} \times 1000 = 27.24 \; per\; thousand$$

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Standardized Death Rate (Indirect Method) =

$$S.D.R = \frac{\sum d_{im} + \sum d_{if}}{\sum (\frac{D_{im}}{P_{im}} \times p_{im}) + \sum (\frac{D_{if}}{P_{if}} \times p_{if})} \times \frac{\sum (D_{im} + D_{if})}{\sum (P_{im} + P_{if})} \times 1000$$

$$= \frac{102 + 85}{79 + 66} \times \frac{1550 + 1280}{70000 + 63000} \times 1000 = 27.44 \; per\; thousand$$

Crude-Birth Rate

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It is a ratio of total registered live births during a specified year to the total midyear population during the same year and multiplied by 1,000.

$$C.B.R = \frac{B}{P} \times 1000$$

Where C.B.R stands for crude birth rate

B denotes the total number of live births registered during a given year

P denotes the midyear total population during the same year.

The difference between the crude birth rate and the crude death rate presents the crude rate of natural increase of the population per year per thousand people.

$$Crude\ rate\ of\ natural\ increase = \frac{Births - Deaths}{Midyear\ Population} \times 1,000$$

The crude birth rate is sometimes called the "crude live-birth rate" to distinguish it from the "still-birth rate" which is computed as follows:

$$Still - birth\ Rate = \frac{B_s}{B} \times 1,000$$

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Where B_s denotes the number of still-births during any year.

B denotes the total number of births in the same year.

Age-Specific Birth Rate

While calculating crude birth rate, we have taken into consideration the total midyear population, but the number of births depends upon the number of married women of child-bearing age. Moreover, births vary with the age of the parents, etc. Age-specific birth rates (A.S.B.R) are computed for women by the formula

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$$C.B.R = \frac{b_i}{P_{if}} \times 1000$$

b_i denotes the number of births registered during the year to women of ith age-group

P_{if} denotes the mid-year population figure of women in the same age-group.

Standardized Birth Rate

- The crude birth rates cannot be used for inter-area comparison because the number of births in any area depends upon the number of married women between the child-bearing ages.
- The crude birth rate for an area containing a very large proportion of population outside the child-bearing ages, is likely to be smaller in spite of the fact that the number of children per woman in the area is greater.
- The standardized birth rate is computed in a manner similar to the standardized death rate by taking a standard population of women between the child-bearing ages given according to different age-groups.

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Example

Calculate age-sex specific birth rates for the following data for selected age-groups.

Age-group (years)	**Registered Births**	**Female Population**
15 – 19	7,890	66,000

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20 – 24	17,760	52,930
25 – 29	15,620	43,520
30 – 34	11,315	38,600
35 – 39	7,670	34,240
40 – 44	2,650	26,670

Age Specific Birth Rate

Age-group (years)	**Registered Births**	**Female Population**	**Age-specific Birth rate per 1000**
15 – 19	7,890	66,000	119.5455
20 – 24	17,760	52,930	335.5375
25 – 29	15,620	43,520	358.9154
30 – 34	11,315	38,600	293.1347
35 – 39	7,670	34,240	224.007
40 – 44	2,650	26,670	99.36258

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Lecture 14

- Observed Fertility Rates
- Total Fertility Rate
- Reproduction Rate
- Vital Index

General Fertility Rate

- Fertility means actual production of children. It is defined as the number of births per 1000 women of child-bearing age (usually taken as 15 – 44 years or sometimes 15 – 49 years). Symbolically,

$$G.F.R = \frac{B}{P_{f(15–44)}} \times 1000$$

- Where B = number of live-births registered during the year.
- $P_{f(15–44)}$ = mid-year female population of child-bearing age

  (15 – 44 years or sometimes 15 – 49 years)

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Example

Suppose the number of registered live-births in a city during 1960 were 63,800. If the female population in the child-bearing age (15 – 44 years) was 261,960 on July 1,1960, then the general fertility rate will be:

$$G.F.R = \frac{B}{P_{f(15–44)}} \times 1000$$

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$$G.F.R = \frac{63,800}{261,960} \times 1000$$

=243.5 per thousand.

Specific Fertility Rate

General fertility rate shows rate of production of children by women of child-bearing age during a year. Age-specific fertility rate refers to the number of females produced by 1000 women of given age-group. The age group are usually 5-yearly age-groups.

$$\text{S}.F.R = h\left(\frac{b_{if}}{P_{if}}\right) \times 1000$$

- Where b_{if} = female births to women of ith age-group
- P_{if} = women of ith age-group
- h = length of the interval which is usually taken as 5.

Total fertility Rate

Total fertility rate is obtained by aggregating the age-specific fertility rates for women of each reproductive age. When the age-specific fertility rate are given or computed 5-year age-groups the aggregate is to be multiplied by 5, the number of years in each age-group, as it is the sum of the rates at every individual age which is required. Total fertility rate is computed by the formula

$$T.F.R = 5\sum \left(age - specific\ fertility\ rate\right)$$

$$= 5\sum_{i} \frac{B_i}{P_{if}} \times 1000$$

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- Where B = number of live-births registered during the year.
- P_{if} = mid-year female population of child-bearing age

  (15 – 44 years or sometimes 15 – 49 years)
- The total fertility rate thus provides the total number of babies who would be born (ignoring mortality) to a hypothetical group of fertility rates while passing through the reproductive period.

Example

Calculate the general fertility rate and total fertility rates from the following data:

Age-Group	Female Population (000)	Births
15 – 19	18	450
20 – 24	16	1760
25 – 29	15	2250
30 – 34	13	1560
35 – 39	10	900
40 – 44	8	320
45 – 49	5	80

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Age-Group	Female Population	Births	$\frac{B_i}{P_{if}} \times 1000$
15 – 19	18000	450	25
20 – 24	16000	1760	110
25 – 29	15000	2250	150
30 – 34	13000	1560	120
35 – 39	10000	900	90
40 – 44	8000	320	40
45 – 49	5000	80	16
Total	85000	7320	551

$$T.F.R = 5\sum\left(age-specific\ fertility\ rate\right)$$

$$= 5\sum_i \frac{B_i}{P_{if}} \times 1000$$

$$= 5(551) = 2,755$$

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$$G.F.R = \frac{B}{P_{f(15-49)}} \times 1000$$

$$= \frac{7320}{85000} \times 1000 = 86.12$$

Reproduction Rate

- The total fertility rate shows the mean number of children which a female is expected to produce during her child-bearing age.
- A more significant rate is the reproduction rate which will give an indication of the number of females which a female will produced over her child-bearing age to replace herself.

Gross Reproduction Rate

- The gross reproduction rate can be considered as a special case of fertility rate restricted to females.
- It measures the mean number of female children which will be born to a newly born female who will be subject to given fertility conditions throughout her reproductive or child-bearing age but not subject to mortality.
- The gross reproduction rate (G.R.R.) can simply be calculated by multiplying the total fertility rate by the proportion of all births which are female, i.e.

$$G.R.R. = \frac{Number\ of\ female\ births}{Number\ of\ births} \times T.F.R$$

- The total fertility rate of Australia during 1960 was 3,452 per women (3,452 per thousand women) and the number of registered births during the same year were 230,326 out of which 111,911 were females. The gross reproduction rate for Australia for 1960 was

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$$G.R.R. = \frac{111,911}{230,326} \times 3,452 = 1.677$$

- If data are available on the number of female births by age of mother, the gross reproduction rate can be calculated directly by adding the age-specific fertility rates. Usually the female population in the child-bearing age is classified by 5-yearly age-groups. In such a case, we can compute the gross reproduction rate by multiplying the sum of age-specific fertility rate by 5, i.e.

$$G.R.R. = 5\sum_{i=1}^{7} \frac{b_{if}}{P_{if}}$$

- Where b_{if} = number of live female births registered during a year to mother ofith age-group
- P_{if} = mid-year female population of the same age-group.
- The gross reproduction rate assumes that (i) all females will survive to the end of the child-bearing age and (ii) a uniform fertility rate throughout the reproductive age, i.e. the fertility rate would remain unchanged throughout the child-bearing age.

Net Reproduction Rate

- It is known fact that all the female babies born do not survive till they reach the child-bearing age.
- To adjust the rate for mortality, we therefore take into account the survival rates (i.e. the probability of daughters surviving from birth to age-group of mother) for girls born to women in different age-groups.
- N.R.R is defined by the average number of female babies who would become mothers when they attain their child-bearing age. It is computed by the formula

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$$N.R.R = 5\sum \frac{Female\,births}{Female\,population} \times probability\,of\,survival.$$

- Hence the N.R.R estimates the number of female babies that would be produced by women throughout their life time if they were to experience at each age-group certain fertility and mortality rates. Thus it estimates the average number of potential mother who survive.
- A net reproduction rate of 1 signifies that the present female population is exactly maintaining itself and the population is considered stable.
- If it is less than 1, it implies that the number of potential mother is decreasing and hence the population is declining.
- If it is greater than 1, it implies that the number of potential mother is increasing and hence the population will increase.

Example

Calculate the gross and net reproduction rates from the following data:

Age-Group	Female Population (000)	Female Births (00)	Survival Rate
15 – 19	1,6000	190	0.921
20 – 24	1,000	702	0.901
25 – 29	1,685	906	0.885
30 – 34	1,730	624	0.862

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35 – 39	1,725	325	0.850
40 – 44	1,620	110	0.832
45 – 49	1,510	8	0.810

Age-Group	Female Population	Female Births	Survival Rate	$\frac{b_{if}}{P_{if}}$	$(\frac{b_{if}}{P_{if}}) \times SR$
15 – 19	16000000	19000	0.921	0.001188	0.001094
20 – 24	1000000	70200	0.901	0.0702	0.06325
25 – 29	1685000	90600	0.885	0.053769	0.047585
30 – 34	1730000	62400	0.862	0.036069	0.031092
35 – 39	1725000	32500	0.850	0.018841	0.016014
40 – 44	1620000	11000	0.832	0.00679	0.005649
45 – 49	1510000	800	0.810	0.00053	0.000429

$$G.R.R = 5\sum_{i=1}^{7} \frac{b_{if}}{P_{if}}$$

$$= 5(0.187386) = 0.93693$$

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$$N.R.R = 5\sum \frac{Female\,births}{Female\,population} \times probability\,of\,survival.$$

$$= 5(0.165114) = 0.82557$$

Example

Calculate the net reproduction rate from the following data:

Age-Group	Female Population	Female Births	Survival Rate
15 – 19	87	4	0.850
20 – 24	63	11	0.800
25 – 29	55	8	0.700
30 – 34	41	6	0.650
35 – 39	33	4	0.600
40 – 44	36	2	0.500

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How would you interpret the result?

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Age-Group	Female Population	Female Births	Survival Rate	
15 – 19	87	4	0.850	0.03908
20 – 24	63	11	0.800	0.139683
25 – 29	55	8	0.700	0.101818
30 – 34	41	6	0.650	0.095122
35 – 39	33	4	0.600	0.072727
40 – 44	36	2	0.500	0.027778
Total				0.476208

$$N.R.R = 5(0.476208) = 2.38104$$

Since the value of N.R.R is greater 1 which shows that the population is increasing.

Example

Calculate general fertility rate, total fertility rate, gross reproduction rate and net reproduction rate from the following data:

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Age-Group	Female Population (000)	Female Births	Number of Survivors out of 10000
15 – 19	1424	27639	9645
20 – 24	1531	226817	9607
25 – 29	1653	280506	9554
30 – 34	1658	194526	9489
35 – 39	1741	113966	9416
40 – 44	1669	32363	9324
45 – 49	1561	2215	9201

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General Fertility Rate

Age-Group	Female Population (000)	Female Births
15 – 19	1424000	27639
20 – 24	1531000	226817

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25 – 29	1653000	280506
30 – 34	1658000	194526
35 – 39	1741000	113966
40 – 44	1669000	32363
45 – 49	1561000	2215
Total	11237000	878032

$$G.F.R = \frac{B}{P_{f(15-49)}} \times 1000$$

$$= \frac{878032}{11237000} \times 1000 = 78.13758$$

Total Fertility Rate

Age-Group	Female Population (000)	Female Births	
15 – 19	1424000	27639	19.40941
20 – 24	1531000	226817	148.1496
25 – 29	1653000	280506	169.6951

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30 – 34	1658000	194526	117.3257
35 – 39	1741000	113966	65.46008
40 – 44	1669000	32363	19.39065
45 – 49	1561000	2215	1.418962
Total			540.8495

$$T.F.R = 5\sum(age-specific\ fertility\ rate)$$

$$= 5\sum_i \frac{B_i}{P_{if}} \times 1000$$

$$= 5(540.895) = 2704.247$$

Gross Reproduction Rate

Age-Group	Female Population (000)	Female Births	
15 – 19	1424000	27639	0.019409
20 – 24	1531000	226817	0.14815

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25 – 29	1653000	280506	0.169695
30 – 34	1658000	194526	0.117326
35 – 39	1741000	113966	0.06546
40 – 44	1669000	32363	0.019391
45 – 49	1561000	2215	0.001419
Total			0.540849

$$G.R.R = 5\sum_{i=1}^{7} \frac{b_{if}}{P_{if}}$$

$$= 5(0.540849) = 2.704247$$

Net Reproduction Rate

Age-Group	Female Population (000)	Female Births	Survival Rate	$(\frac{b_{if}}{P_{if}}) \times SR$
15 – 19	1424000	27639	0.9645	0.01872
20 – 24	1531000	226817	0.9607	0.142327
25 – 29	1653000	280506	0.9554	0.162127

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30 – 34	1658000	194526	0.9489	0.11133
35 – 39	1741000	113966	0.9416	0.061637
40 – 44	1669000	32363	0.9324	0.01808
45 – 49	1561000	2215	0.9201	0.001306
Total				0.515527

$$N.R.R = 5(0.5155274) = 2.577637$$

Vital Index

- The vital index is the ratio of the number of births (B) to the number of deaths (D) multiplied by 100. Thus

$$V.I = \frac{B}{D} \times 100$$

- This index indicates the extent to which the force of natality exceeds the force of mortality at a given time.

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Lecture 15

Example

Give the following data for the two districts:

Age Groups	District A		District B	
	Population	Deaths	Population	Deaths
0 – 14	500	2	400	1
15 – 29	2000	8	300	1
30 – 44	2000	12	100	5
45 – 59	1000	10	2000	18
60 – 74	500	20	2000	70
75 – 89	100	15	400	50

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Compute the crude death rates for districts A and B.

Age Groups	District A		District B	
	Population	Deaths	Population	Deaths
0 – 14	500	2	400	1
15 – 29	2000	8	300	1
30 – 44	2000	12	100	5

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45 – 59	1000	10	2000	18
60 – 74	500	20	2000	70
75 – 89	100	15	400	50

$$C.D.R\ for\ District\ A = \frac{Deaths\ of\ District\ A}{Population\ of\ District\ A} \times 1000$$

$$= \frac{67}{6100} \times 1000 = 10.98$$

$$C.D.R\ for\ District\ B = \frac{Deaths\ of\ District\ B}{Population\ of\ District\ B} \times 1000$$

$$= \frac{145}{5200} \times 1000 = 27.88$$

Example

Compute specific mortality rates from cancer per 100,000 population from the following data:

Age Groups	Population as of census data	Deaths from Cancer
20 – 29	725,369	79
30 – 39	700,213	238
40 – 49	609,616	660
50 – 59	548,400	1,582

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60 – 69	402,054	2,342
70 and above	237,280	3,402

$$A.S.D.R = \frac{d_i}{p_i} \times 1000$$

Age Groups	Population as of census data	Deaths from Cancer	Age Specific Mortality Rate
20 – 29	725,369	79	0.000109
30 – 39	700,213	238	0.00034
40 – 49	609,616	660	0.001083
50 – 59	548,400	1,582	0.002885
60 – 69	402,054	2,342	0.005825
70 and above	237,280	3,402	0.014337

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Example

Calculate age-specific death rate per 1000 persons of Pakistan from the following data:

Age (in completed years)	1964 Estimated Population (000)	1964 Estimated Deaths (00)

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Under 1	1,363	3,115
1 – 4	5,578	1,841
5 – 9	6,595	401
10 – 19	8,421	255
20 – 29	6,818	329
30 – 39	5,150	309
40 – 49	3,686	316
50 - 59	2,403	277

$$A.S.D.R = \frac{d_i}{p_i} \times 1000$$

Age (in completed years)	1964 Estimated Population (000)	1964 Estimated Deaths (00)	A.S.D.R
Under 1	1363000	311500	228.54
1 – 4	5578000	184100	33.00466
5 – 9	6595000	40100	6.080364
10 – 19	8421000	25500	3.028144

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20 – 29	6818000	32900	4.825462
30 – 39	5150000	30900	6
40 – 49	3686000	31600	8.572979
50 - 59	2403000	27700	11.52726

Example

Calculate the crude and standardized death rates of the local population from the following data using direct method.

Age-group (years)	Standard Population	No. of Deaths	Local Population	No. of Deaths
0 – 10	600	18	400	16
10 – 20	1000	5	1500	6
20 – 60	3000	24	2400	24
60 & over	400	20	700	21

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$$C.D.R\ for\ local\ Population = \frac{D}{P} \times 1000$$

$$= \frac{67}{5000} \times 1000 = 13.4\ per\ thousand$$

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$$S.D.R = \frac{\sum\left(\frac{d_i}{p_i} \times P_i\right)}{\sum P_i} \times 1000$$

$$Expected\ Death\ Rate = \frac{d_i}{p_i} \times P_i$$

Age-group (years)	Standard Population	Expected Death in Standard Population
0 – 10	600	24
10 – 20	1000	4
20 – 60	3000	30
60 & over	400	12

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$$S.D.R\ for\ local\ Population = \frac{\sum\left(\frac{d_i}{p_i} \times P_i\right)}{\sum P_i} \times 1000$$

$$= \frac{70}{5000} \times 1000 = 14\ per\ thousand$$

Example

Calculate standardized death rate by indirect method from the following data:

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Age-Group	Standard Population	Deaths	Locality Population
0 – 1	9001	83	901
1 – 5	25957	304	1234
5 – 15	12918	181	3034
15 – 30	17076	164	4419
30 – 50	7831	60	799
50 & over	3601	122	263

Given total of registered deaths in locality is 338

$$S.D.R = \frac{\sum d_i}{\sum \left(\frac{D_i}{P_i} \times p_i \right)} \times C.D.R$$

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$$Expected\ Death\ Rate = \frac{D_i}{P_i} \times p_i$$

Age-Group	Standard Population	Deaths	Locality Population	Expected Death Rate
0 – 1	9001	83	901	8.308299
1 – 5	25957	304	1234	14.45221

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5 – 15	12918	181	3034	42.51076
15 – 30	17076	164	4419	42.44062
30 – 50	7831	60	799	6.121824
50 & over	3601	122	263	8.910303

Example

Calculate the standardized death rates for District A, taking the population of the Province as standard from the following data using

1) Direct Method 2) Indirect Method

Age-group (years)	Population of the Province		Death in the Province		Population of District A		Deaths in District A	
	Male	Female	Male	Female	Male	Female	Male	Female
0 – 1	2431	2391	483	394	123	116	36	28
1 – 5	2316	7854	272	245	445	427	15	13
5 – 10	9385	8032	103	93	513	450	4	5
10 – 15	8405	6785	86	75	444	372	3	3
15 – 20	6352	5158	75	63	303	272	2	2

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20 – 30	12401	12304	123	115	582	530	5	6
30 - 40	9202	7130	120	105	436	370	6	7

Standardized Death Rate (Direct Method) =

$$S.D.R = \frac{\sum \frac{d_{im}}{p_{im}} P_{im} + \sum \frac{d_{if}}{p_{if}} P_{if}}{\sum (P_{im} + P_{if})} \times 1000$$

Standardized Death Rate (Indirect Method) =

$$S.D.R = \frac{\sum d_{im} + \sum d_{if}}{\sum (\frac{D_{im}}{P_{im}} \times p_{im}) + \sum (\frac{D_{if}}{P_{if}} \times p_{if})}$$

Age-group (years)	Expected Death in Standard Population		Expected deaths in District A on the basic of deaths in Standard Population (Province)	
	Male	Female	Male	Female
	$\frac{d_{im}}{p_{im}} \times P_{im}$	$\frac{d_{if}}{p_{if}} \times P_{if}$	$\frac{D_{im}}{P_{im}} \times p_{im}$	$\frac{D_{if}}{P_{if}} \times p_{if}$
0 – 1	711.5122	577.1379	24.43809	19.11501
1 – 5	78.06742	239.1148	52.26252	13.31996

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5 – 10	73.17739	89.24444	5.630155	5.210408
10 – 15	56.79054	54.71774	4.54301	4.112012
15 – 20	41.92739	37.92647	3.577613	3.322218
20 – 30	106.5378	139.2906	5.772599	4.953674
30 – 40	126.633	134.8919	5.68572	5.448808
Total	1194.646	1272.324	101.9097	55.4821

$$S.D.R = \frac{\sum \frac{d_{im}}{p_{im}} P_{im} + \sum \frac{d_{if}}{p_{if}} P_{if}}{\sum (P_{im} + P_{if})} \times 1000$$

$$= \frac{1194.646 + 1272.324}{50492 + 49654} \times 1000 = 24.63 \; per \; thousand$$

Standardized Death Rate (Indirect Method)

$$= \frac{\sum d_{im} + \sum d_{if}}{\sum (\frac{D_{im}}{P_{im}} \times p_{im}) + \sum (\frac{D_{if}}{P_{if}} \times p_{if})} \times \frac{\sum (D_{im} + D_{if})}{\sum (P_{im} + P_{if})} \times 1000$$

$$= \frac{71 + 64}{101.9097 + 55.4821} \times \frac{1262 + 1090}{50492 + 49654} \times 1000$$

= 20.144 per thousand.

Example

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Calculate the general fertility rate, total fertility rate and gross reproduction rate and net reproduction rate from the following data:

Age-Group	Female Population	Numbers of births	Survivor Ratio
15 – 19	4477944	421	0.894
20 – 24	3028611	108134	0.786
25 – 29	2356296	358700	0.745
30 – 34	3158974	348560	0.732
35 – 39	2927168	210925	0.712
40 – 44	2699471	91889	0.691
45 – 49	2520514	23899	0.688

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General Fertility Rate

Age-Group	Female Population	Numbers of births
15 – 19	4477944	421
20 – 24	3028611	108134
25 – 29	2356296	358700

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30 – 34	3158974	348560
35 – 39	2927168	210925
40 – 44	2699471	91889
45 – 49	2520514	23899
Total	21168978	1142528

$$G.F.R = \frac{B}{P_{f(15-49)}} \times 1000$$

$$= \frac{1142528}{21168978} \times 1000$$

$$= 53.97 \; per \; thousand$$

Total Fertility Rate

Age-Group	Female Population (000)	Female Births	$\frac{B_i}{P_{if}} \times 1000$
15 – 19	4477944	421	0.094016
20 – 24	3028611	108134	35.70416
25 – 29	2356296	358700	152.2304

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30 – 34	3158974	348560	110.3396
35 – 39	2927168	210925	72.0577
40 – 44	2699471	91889	34.03963
45 – 49	2520514	23899	9.481796
Total	21168978	1142528	413.9474

$$= 5(413.9474) = 2069.737 = 2070$$

Gross Reproduction Rate

Age-Group	Female Population (000)	Female Births	$\frac{b_{if}}{P_{if}}$
15 – 19	4477944	421	9.40164E-05
20 – 24	3028611	108134	0.035704156
25 – 29	2356296	358700	0.15223045
30 – 34	3158974	348560	0.110339623
35 – 39	2927168	210925	0.072057702
40 – 44	2699471	91889	0.034039632

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45 – 49	2520514	23899	0.009481796
Total	21168978	1142528	0.413947376

$$G.R.R = 5\sum_{i=1}^{7} \frac{b_{if}}{P_{if}}$$

$$= 5(0.41395) = 2.06975 = 2.070$$

Net Reproduction Rate

Age-Group	Female Population (000)	Female Births	Survival Rate	$(\frac{b_{if}}{P_{if}}) \times SR$
15 – 19	4477944	421	0.894	8.40506E-05
20 – 24	3028611	108134	0.786	0.028063467
25 – 29	2356296	358700	0.745	0.113411685
30 – 34	3158974	348560	0.732	0.080768604
35 – 39	2927168	210925	0.712	0.051305084
40 – 44	2699471	91889	0.691	0.023521386
45 – 49	2520514	23899	0.688	0.006523476
Total	21168978	1142528		0.303677752

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$$_{s} = 5(0.3037) = 1.519$$

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